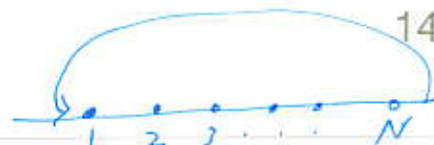


1D Ising chain exact solution

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$$E = -J \sum_{i=1}^N \sigma_i \sigma_{(i+1) \bmod N} - h \sum_{i=1}^N \sigma_i$$



Date

$$= -J \sum_{i=1}^N \left(J \sigma_i \sigma_{i+1} + \frac{h}{2} (\sigma_i + \sigma_{i+1}) \right) = - \sum_{i=1}^N E(\sigma_i, \sigma_{i+1})$$

$N+1$ is same as 1



$$Z = \sum_{\sigma_1} \sum_{\sigma_2} \dots \sum_{\sigma_N} e^{\sum_{i=1}^N \left(J \sigma_i \sigma_{i+1} + \frac{h}{2} (\sigma_i + \sigma_{i+1}) \right)}$$

$$E(\sigma_i, \sigma_{i+1}) =$$

$$J(\sigma_i, \sigma_{i+1}) + \frac{h}{2} (\sigma_i + \sigma_{i+1})$$

$$= \sum_{\sigma_1} \sum_{\sigma_2} \dots \sum_{\sigma_N} e^{\beta E(\sigma_1, \sigma_2)} e^{\beta E(\sigma_2, \sigma_3)} e^{\beta E(\sigma_3, \sigma_4)} \dots e^{\beta E(\sigma_N, \sigma_1)}$$

define $P(\sigma_i, \sigma_{i+1}) = e^{\beta E(\sigma_i, \sigma_{i+1})} = P_{\sigma_i, \sigma_{i+1}}$

$$P = \begin{matrix} \sigma_i = +1 & \sigma_{i+1} = +1 \\ \sigma_i = +1 & \sigma_{i+1} = -1 \\ \sigma_i = -1 & \sigma_{i+1} = +1 \\ \sigma_i = -1 & \sigma_{i+1} = -1 \end{matrix} \begin{pmatrix} e^{\beta(J+h)} & e^{-\beta J} \\ e^{-\beta J} & e^{\beta(J-h)} \end{pmatrix}$$

$$(AB)_{ij} = \sum_k A_{ik} B_{kj}$$

$$= \sum_{\sigma_1, \sigma_2, \dots, \sigma_N} P_{\sigma_1, \sigma_2} P_{\sigma_2, \sigma_3} P_{\sigma_3, \sigma_4} \dots P_{\sigma_N, \sigma_1}$$

$$= \text{Tr } P \cdot \underbrace{P \dots P}_N = \text{Tr } P^N = \text{Tr } S P S^{-1} S P S^{-1} S P \dots P$$

$$S P S^{-1} = \begin{pmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{pmatrix}$$

$$= \lambda_1^N + \lambda_2^N$$

$$\det \left(\det (P - \lambda I) \right) = \det \begin{pmatrix} e^{\beta(J+h)} - \lambda & e^{-\beta J} \\ e^{-\beta J} & e^{\beta(J-h)} - \lambda \end{pmatrix} = 0$$

$$\det \begin{pmatrix} a/b - \lambda & 1/a \\ 1/a & a/b - \lambda \end{pmatrix} = 0$$

Let $a = e^{\beta J}$
 $b = e^{\beta h}$

$$\sinh^2 x - \cosh^2 x = -1$$

$$\frac{e^{2x} - 2 + e^{-2x}}{4} - \frac{e^{2x} + 2 + e^{-2x}}{4}$$

$$\frac{e^{\beta J} \sinh(\beta h) + e^{\beta J} \frac{\sinh(\beta h) \cosh(\beta h)}{\sqrt{\sinh^2(\beta h) + e^{-4\beta J}}}}{e^{\beta J} \cosh(\beta h) + e^{\beta J} \sqrt{\sinh^2(\beta h) + e^{-4\beta J}}}}$$

$$= \frac{\sinh(\beta h) \sqrt{\sinh^2(\beta h) + e^{-4\beta J}} + \sinh(\beta h) \cosh(\beta h)}{\cosh(\beta h) \sqrt{\sinh^2(\beta h) + e^{-4\beta J}} + \sinh^2(\beta h) + e^{-4\beta J}}$$

$$(a \cdot b - \lambda) \left(\frac{a}{b} - \lambda \right) - \left(\frac{1}{a} \right)^2 = 0$$

$$a^2 - (ab + \frac{a}{b})\lambda + \lambda^2 - \frac{1}{a^2} = 0$$

$$\lambda^2 - a(b + \frac{1}{b})\lambda + a^2 - \frac{1}{a^2} = 0$$

$$\lambda_{\pm} = \frac{1}{2} \left(a(b + \frac{1}{b}) \pm \sqrt{a^2(b + \frac{1}{b})^2 - 4(a^2 - \frac{1}{a^2})} \right)$$

$$= e^{\beta J} \left(\frac{e^{\beta h} + e^{-\beta h}}{2} \right) \pm \sqrt{e^{2\beta J} \left(\frac{e^{\beta h} + e^{-\beta h}}{2} \right)^2 - (e^{2\beta J} - e^{-2\beta J})}$$

$$= e^{\beta J} \cosh(\beta h) \pm \sqrt{e^{2\beta J} \cosh^2(\beta h) - e^{2\beta J} + e^{-2\beta J}}$$

$\uparrow$
 $\sinh^2(\beta h) + 1$

$$= e^{\beta J} \cosh(\beta h) \pm \sqrt{e^{2\beta J} \sinh^2(\beta h) + e^{-2\beta J}}$$

$$\lambda_+ > \lambda_-$$

$$F = -k_B T \ln (\lambda_1^N + \lambda_2^N) \quad N \rightarrow \infty$$

$$= -k_B T \ln \lambda_1^N \left(1 + \left(\frac{\lambda_2}{\lambda_1} \right)^N \right)$$

$$\rightarrow 0 \text{ as } N \rightarrow \infty$$

$$= -N k_B T \ln \lambda_1$$

magnetization

$$M = - \frac{\partial F}{\partial h} = - \frac{\partial}{\partial h} \left(- \frac{N}{\beta} \ln \lambda_1 \right)$$

$$= N \frac{1}{\beta \lambda_1} \frac{\partial}{\partial h} \lambda_1 = \frac{N \beta}{\lambda_1} \left[e^{\beta J} \sinh(\beta h) + \frac{2 e^{2\beta J} \sinh(\beta h) \cosh(\beta h)}{2 \sqrt{e^{2\beta J} \sinh^2(\beta h) + e^{-2\beta J}}} \right]$$

$$= N \frac{\sinh \beta h}{\sqrt{\sinh^2(\beta h) + e^{-4\beta J}}}$$

$$J=0, \text{ we get}$$

$$M = N \tanh(\beta h)$$

$$e^{\beta J \sigma_i \sigma_{i+1}} = \cosh(\beta J) + \sigma_i \sigma_{i+1} \sinh(\beta J)$$

$$Z = \sum_{\{\sigma\}} \prod_{i=1}^N \cosh(\beta J) [1 + \sigma_i \sigma_{i+1} \tanh(\beta J)] \quad \sigma_i^2 = 1$$

$$= \cosh^N(\beta J) \sum_{\{\sigma\}} \left[1 + \underbrace{\sigma_1 \sigma_2 \sigma_2 \sigma_3 \cdots \sigma_{N-1} \sigma_N \sigma_N \sigma_1}_{1} \tanh^N(\beta J) \right]$$

$$= 2^N \cosh^N(\beta J) (1 + \tanh^N(\beta J))$$

$$\tanh x = \frac{e^x - e^{-x}}{e^x + e^{-x}} = \frac{1 - e^{-2x}}{1 + e^{-2x}} \\ \approx (1 - e^{-2x})^2 \\ = 1 - 2e^{-2x}$$

$$\lambda_+ = e^{\beta J} \cosh(\beta h) + e^{\beta J} \sqrt{\sinh^2(\beta h) + e^{-4\beta J}}$$

$$\begin{aligned} \frac{\partial \ln \lambda_+}{\partial h} &= \frac{1}{\lambda_+} \frac{\partial \lambda_+}{\partial h} = \frac{e^{\beta J} \left[\sinh(\beta h) + \frac{1}{2} \frac{2 \sinh(\beta h) \cosh(\beta h)}{\sqrt{\sinh^2(\beta h) + e^{-4\beta J}}} \right]}{e^{\beta J} (\cosh(\beta h) + \sqrt{\sinh^2(\beta h) + e^{-4\beta J}})} \\ &= \frac{\sinh(\beta h)}{\sqrt{\sinh^2(\beta h) + e^{-4\beta J}} + \cosh(\beta h)} \\ &= \frac{\sinh(\beta h)}{\sqrt{\sinh^2(\beta h) + e^{-4\beta J}}} \end{aligned}$$

pair correlation function

$\lim_{h \rightarrow 0} M \rightarrow 0$
 $h \rightarrow 0$
 $\lim_{h \rightarrow 0} M = 1$
 $h \rightarrow 0$
 $h = 0$ for simplicity

$$g(j) = \langle \sigma_0 \sigma_j \rangle = \langle \sigma_i \sigma_{i+j} \rangle$$

$$= \sum_{\{\sigma\}} \sigma_0 \sigma_j e^{\beta J (\sigma_0 \sigma_1 + \sigma_1 \sigma_2 + \dots + \sigma_{N-1} \sigma_0)} \frac{1}{Z}$$

$$\tanh x = \frac{\sinh x}{\cosh x}$$

$$e^{\beta J \sigma_1 \sigma_2} = \cosh(\beta J) [1 + \sigma_1 \sigma_2 \tanh(\beta J)] \quad \left(\frac{\lambda}{\lambda} \right) \times$$

$$= \frac{\sum_{\{\sigma\}} \sigma_0 \sigma_j \prod_{i=0}^{N-1} (1 + \sigma_i \sigma_{i+1} \tanh(\beta J))}{\sum_{\{\sigma\}} \prod_{i=0}^{N-1} (1 + \sigma_i \sigma_{i+1} \tanh(\beta J))}$$

$N \rightarrow \infty$

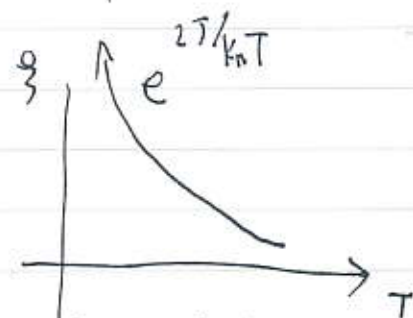
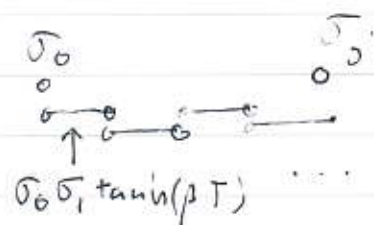
$$= \frac{1}{1} (\tanh^j(\beta J))$$

$$= e^{-\frac{j}{\xi}}$$

$$\tanh(\beta J) = e^{-\frac{1}{\xi}}$$

$$\xi = - \frac{1}{\ln \tanh(\beta J)} \approx \frac{1}{2} e^{2\beta J} \quad \beta \rightarrow \infty$$

meaning ξ



typical size of domains

+++++ - - - - -
 $\rightarrow \xi \leftarrow$

1 (a) energy U ,
volume V
particle number N

Date

(b) Postulate II

(c) monotonicity $\frac{\partial S}{\partial U} \geq 0$

extensivity $S(\lambda U, \lambda V, \lambda N) = \lambda S(U, V, N)$

concavity $\frac{\partial^2 S}{\partial U^2} < 0$ or $2S(U) \geq S(U-\delta U) + S(U+\delta U)$

$$S = 0 \text{ when } T = \frac{\partial U}{\partial S} = 0$$



(d) $\Delta Q = T \Delta S$

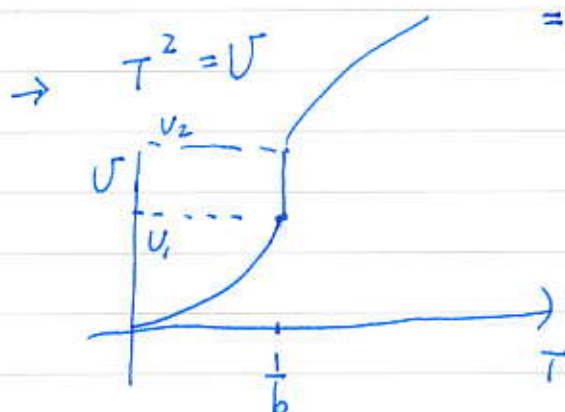
$$= \frac{1}{b} (S_2 - S_1)$$

(b) $\frac{1}{T} = \frac{\partial S}{\partial U} = \begin{cases} \frac{1}{2} a \sqrt{\frac{N}{U}} & U < U_1 \\ b & U_1 \leq U \leq U_2 \\ \frac{1}{2} a' \sqrt{\frac{N}{U}} + d & U_2 < U \end{cases}$

$$= \frac{1}{b} (a' \sqrt{U_2 N} + d U_2 - a \sqrt{U_1 N})$$

$$= \frac{1}{b} (b U_2 + c - b U_1 - c) = (U_2 - U_1)$$

$$T = \begin{cases} \frac{2}{a} \sqrt{\frac{U}{N}} \\ \frac{1}{b} \\ \frac{1}{\frac{a'}{2} \sqrt{\frac{N}{U}} + d} \end{cases}$$



(c) $\begin{cases} a \sqrt{U_1 N} = b U_1 + c \\ b U_2 + c = a' \sqrt{U_2 N} + d U_2 \\ \frac{1}{2} a \sqrt{\frac{N}{U_1}} = b \\ b = \frac{1}{2} a' \sqrt{\frac{N}{U_2}} + d \end{cases}$

(e) $U = \lambda \frac{U_1}{N} + (1-\lambda) \frac{U_2}{N}$

$$S = \lambda S_1 + (1-\lambda) S_2$$

$$0 \leq \lambda \leq 1$$

$\boxed{U_1 \quad U_2}$ fraction

see C. Domb, "The Critical Point" chap 14
 also J.M. Yeomans, or Kubo, Tada, Hashitsume
 Tada, Kubo, Santo

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2D Ising model

$$E = -J \sum_{\langle ij \rangle} \sigma_i \cdot \sigma_j - h \sum_i \sigma_i$$

Date $h=0$

High-T expansion

$$Z = \sum_{\{\sigma\}} e^{\beta J \sum_{\langle ij \rangle} \sigma_i \cdot \sigma_j} = \sum_{\{\sigma\}} \prod_{\langle ij \rangle} \cosh K \cdot (1 + \sigma_i \cdot \sigma_j \tanh K)$$

$$K = \beta J = \frac{J}{k_B T} = \cosh K \sum_{\{\sigma\}} \prod_{\langle ij \rangle} (1 + \sigma_i \cdot \sigma_j \tanh K)$$

L number of links.

$$= \sum_{\langle ij \rangle} 1 = 2^N \cosh^L K \left(1 + \sum_{r=1}^L p(r) \tanh^r K \right)$$

from $\sum$

$p(r)$ ~~# of polygons that meets with an even # of lines~~
~~closed~~

for square lattice



but not



$p(r)$ the number of figures (graphs) with r lines

composed of closed polygons (polygons can be disconnected)

The lines must meet an even # of times at a site.

Low-T expansion

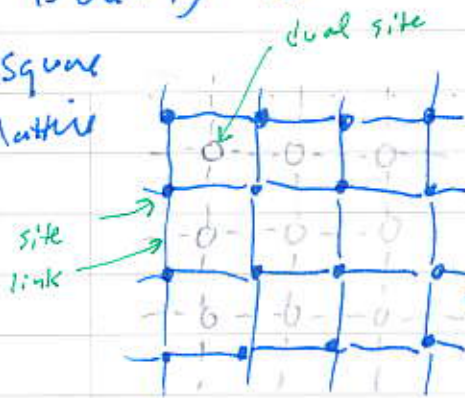
$$Z = \sum_{\{\sigma\}} e^{\beta J \sum_{\langle ij \rangle} \sigma_i \cdot \sigma_j} = 2 \cdot e^{K \cdot L} \left(1 + \sum_{r=1}^{r_{\max}} V(r) e^{-2rk} \right)$$

↑
ground state degeneracy

r : number of + to - boundaries

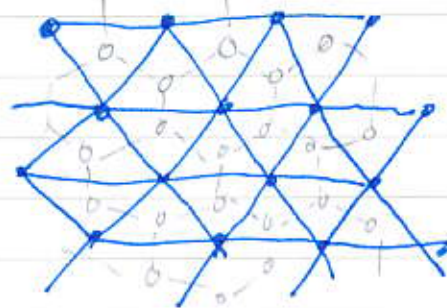
V : number of configs with r + to - boundaries

Duality relation

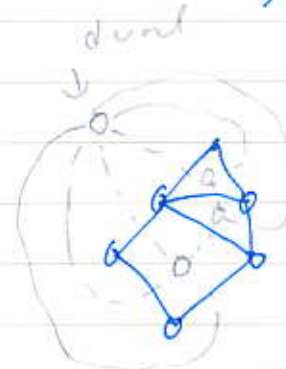
Square
lattice① center of a plaquette is a
dual site

Date

② dual line cross origl line

triangl
latticethe concept works
for any planar
graphs.

finite lattice

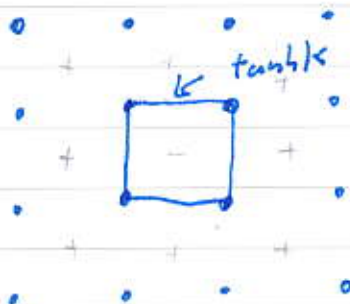


"The number of $P(r)$ for high-temperature expansion
is exactly the same number of $V(r)$ in a low temperature
expansion on the dual lattice"

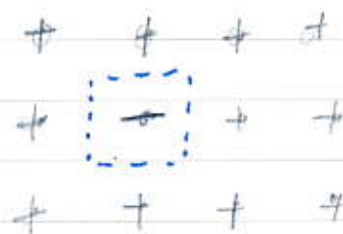
A config (graph) for high-T expansion is also
a config for Low-T expansion on dual lattice
The mapping is one-to-one.

origl lattice

dual lattice



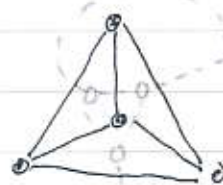
→



let use $*$ to mean the same quantity in dual lattice

$$P(r) = V(r)^*$$

$$V(r) = P(r)^*$$



4-site
Self-dual
finite lattice
(graph or net)

$$Z(N, K) = 2^N (\cosh K)^L \left(\sum_{r=0}^L p(r) \tanh^r K \right) \quad \text{particle fun at each lattice} \quad 161$$

$$Z^*(N^*, K^*) = 2 \exp(K^* L^*) \left(\sum_{r=0}^{r_{\max}^*} p(r) e^{-2K^* r} \right) \quad \text{Date} \quad p(0) = 1$$

$$\frac{Z(N, K)}{2^N (\cosh K)^L} = \frac{Z^*(N^*, K^*)}{2 e^{K^* L^*}} \quad \text{if we choose} \quad \tanh K = e^{-2K^*}$$

for square lattice $Z = Z^*$ because self-dual
 $N = N^*$ $\begin{cases} L = L^* \\ N = N^* \end{cases}$



if $T_c \neq T_c^*$

at T_c f is singular $\rightarrow N \rightarrow \infty$ then 

We would have 2 singular points (two phase transitions).

but if we assume there is only one unique singular pt. then we must have $T_c = T_c^*$ or $\tanh K_c = e^{-2K_c}$

$$K_c \approx 0.4407 = -\ln(0.68679)$$

$$\text{or } e^{-2K_c} = \sqrt{2} - 1$$

$$\text{symmetry for } \sinh(2K) \sinh(2K^*) = 1$$

$$\text{topological relation } L = L^*$$

$$N + N^* = L + 2 \quad \text{Euler relation}$$

$$\# \text{ of site} - \# \text{ of lines} + \# \text{ of faces} = 2$$



$$N = 5$$

$$N^* = 3$$

$$L = 6$$

$$v - e + f = 2$$

valid for planar graphs or simply connected polygons

$$3. \quad \mathcal{H} = \frac{p^2}{2m} + \frac{1}{2} m \omega^2 q^2$$

$$(a) \quad Z = \frac{1}{h} \int dp \int dq e^{-\beta \left(\frac{p^2}{2m} + \frac{1}{2} m \omega^2 q^2 \right)}$$

Date

$$= \frac{1}{h} \int_{-\infty}^{\infty} e^{-\frac{\beta p^2}{2m}} dp \int_{-\infty}^{\infty} e^{-\frac{\beta m \omega^2 q^2}{2}} dq$$

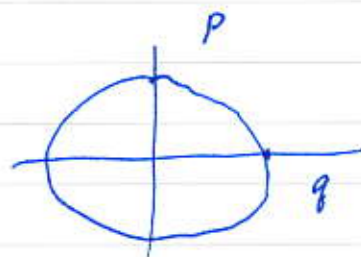
$$= \frac{1}{h} \sqrt{\frac{2\pi T m}{\beta}} \sqrt{\frac{2\pi}{\beta m}} \frac{1}{\omega} = \frac{2\pi}{h \beta \omega} = \frac{2\pi k_B T}{h \omega} = \frac{k_B T}{\hbar \omega}$$

$$F = -k_B T \ln \left(\frac{k_B T}{\hbar \omega} \right)$$

$$S = -\frac{\partial F}{\partial T} = +k_B \ln \left(\frac{k_B T}{\hbar \omega} \right) + k_B$$

$$U = F + TS = k_B T$$

$$(b) \quad \textcircled{1} \quad \Gamma = \int_{H < U} dp dq$$



= area of ellipse

$$= \pi \sqrt{2 \frac{U}{m \omega^2}} \sqrt{2 m U} = \frac{2\pi U}{\omega}$$

$$S = k_B \ln \left(\frac{2\pi U}{\omega} \right)$$

$$\frac{1}{T} = \frac{\partial S}{\partial U} = k_B \frac{1}{U} \quad \text{ie. } U = k_B T$$

$$F = U - TS = k_B T - T k_B \ln \left(\frac{2\pi k_B T}{\omega} \right)$$

$$\textcircled{2} \quad \Omega = \Gamma(U + \Delta) - \Gamma(U) = \frac{2\pi \Delta}{\omega}$$

$$S = k_B \ln \left(\frac{2\pi \Delta}{\omega} \right)$$

$$\frac{1}{T} = \frac{\partial S}{\partial U} = 0 \quad T \rightarrow \infty \quad U \text{ ill-defined}$$

F ill-defined

(c) because we did not take $N \rightarrow \infty$. $N = 1$

(d) $\Omega = \Omega(U, N) = \left(\frac{2\pi U}{\omega}\right)^N \cdot \text{const}$

$$\Omega \approx \Delta \frac{d\Omega}{dU} = \text{const}' \left(\frac{2\pi U}{\omega}\right)^{N-1}$$

In $N \rightarrow \infty$
they are
the same.

const $\frac{\pi^{\frac{n}{2}}}{(\frac{n}{2})!}$

4. (a) $T_e = T_g$

$$P_e = P_g$$

$$\mu_e = \mu_g$$

(b) $S = S_e + S_g$

$$U_e + U_g = U$$

$$0 = dS = \frac{\partial S_e}{\partial U_e} dU_e + \frac{\partial S_e}{\partial V_e} dV_e + \frac{\partial S_e}{\partial N_e} dN_e + \frac{\partial S_g}{\partial U_g} dU_g + \dots$$

$$= \left(\frac{1}{T_e} - \frac{1}{T_g}\right) dU_e + \left(\frac{P_e}{T_e} - \frac{P_g}{T_g}\right) dV_e + (\mu_e - \mu_g) dN_e$$

//

$$T_e = T_g, \quad P_e = P_g, \quad \mu_e = \mu_g$$

(c) $\mu_g = \mu_g$ means

$$\mu_e - \mu_g = 0$$

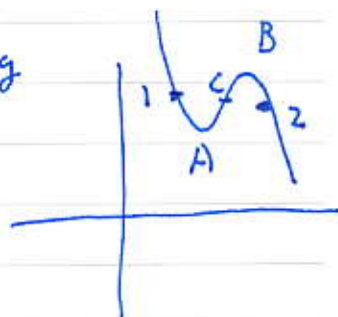
using Gibbs - Duhem relation

on the isotherm curve $dT = 0$

$$0 = \mu_e - \mu_g = \int_{\mu_g}^{\mu_e} d\mu = \int \frac{V}{N} dP = 0$$

$$d\mu = -\frac{S}{N} dT + \frac{V}{N} dP$$

$$\int_{I, A, C, B, 2} V dP = 0 \rightarrow \text{Area I} = \text{Area II}$$



2D Ising basic idea

write the transfer matrix

 $V = e^{\text{quadratic terms of spin/fermion operator}}$

$$H = -J \sum_{i=1}^N \sigma_i \sigma_{i+1} - h \sum_{i=1}^N \sigma_i$$

$$\beta J = K$$

$$Z = \sum_{\{\sigma\}} \underbrace{e^{\beta h \sigma_1}}_{V_1} \underbrace{e^{K \sigma_1 \sigma_2}}_{V_2} \underbrace{e^{\beta h \sigma_2}}_{V_1} \underbrace{e^{K \sigma_2 \sigma_3}}_{V_2} \dots e^{\beta h \sigma_N} \underbrace{e^{K \sigma_N \sigma_1}}_{V_2}$$

$$= \sum \langle \sigma_0 | V_1 | \sigma_1 \rangle \langle \sigma_1 | V_2 | \sigma_2 \rangle \langle \sigma_2 | V_1 | \dots \dots | \sigma_0 \rangle$$

matrix elements

$$\hat{\sigma}_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \quad \hat{\sigma}^+ = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \quad \hat{\sigma}^- = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}$$

$$\hat{\sigma}_x = \hat{\sigma}^+ + \hat{\sigma}^-, \quad \hat{\sigma}_y = -i (\hat{\sigma}^+ - \hat{\sigma}^-)$$

$$\begin{aligned} \langle +1 | V_1 | +1 \rangle &= e^{\beta h} & \langle +1 | V_1 | -1 \rangle &= 0 \\ \langle -1 | V_1 | -1 \rangle &= e^{-\beta h} & \langle -1 | V_1 | +1 \rangle &= 0 \end{aligned}$$

$$\langle +1 | V_2 | +1 \rangle = \langle -1 | V_2 | -1 \rangle = e^K$$

$$\langle +1 | V_2 | -1 \rangle = \langle -1 | V_2 | +1 \rangle = e^{-K}$$

matrix

$$V_1 = e^{\beta h \hat{\sigma}_z} = 1 + e^{\beta h} \hat{\sigma}_z + e^{\beta h} \hat{\sigma}_z^2 + \dots = \begin{pmatrix} e^{\beta h} & 0 \\ 0 & e^{-\beta h} \end{pmatrix}$$

$$V_2 = e^K \mathbb{1} + e^{-K} \sigma_x = A e^{K^* \sigma_x} \quad \sigma_x^2 = 1$$

$$= A \left\{ 1 + K^* \sigma_x + \frac{1}{2} (K^* \sigma_x)^2 + \frac{(K^* \sigma_x)^3}{3!} + \dots \right\}$$

$$= A \left\{ 1 + \frac{1}{2} K^{*2} + \frac{1}{4!} K^{*4} + \dots + \left(K^* + \frac{K^{*3}}{3!} + \dots \right) \sigma_x \right\}$$

$$= A \cosh K^* + A \sinh K^* \sigma_x$$

$$\rightarrow A \cosh K^* = e^K, \quad A \sinh K^* = e^{-K}$$

$$Z = \text{Tr}(V_1 V_2)^N = \text{Tr}(V_2^{\frac{1}{2}} V_1 V_2^{\frac{1}{2}})^N = \lambda_1^N + \lambda_N^N \leftarrow \text{eigenvalues of } V \quad 169$$

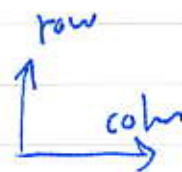
$$V = V_2^{\frac{1}{2}} V_1 V_2^{\frac{1}{2}} = \sqrt{2 \sinh(2K)} e^{K^* \hat{\sigma}_x / 2} e^{\beta h \hat{\sigma}_z / 2} e^{K^* \hat{\sigma}_x / 2}$$

$$\sigma_{ij} \quad \cdot \quad \cdot \quad \cdot \quad \sigma_{M,M}$$

$$\begin{array}{ccc} \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot \end{array}$$

$$\cdot \quad \cdot \quad \cdot$$

$$\sigma_{1,1} \quad \sigma_{1,2} \quad \cdot \quad \sigma_{1,M}$$



$$H = -J \sum_{r,c} \sigma_{r,c} \sigma_{r+1,c} - J \sum_{r,c} \sigma_{r,c} \sigma_{r,c+1}$$

states of a particular column

$$2^M \text{ basis } |\mu\rangle = |\mu_1, \mu_2, \dots, \mu_M\rangle = |\mu_1\rangle |\mu_2\rangle \dots |\mu_M\rangle$$

$$\hat{\sigma}_{j,x} \quad \hat{\sigma}_{j,y} \quad \hat{\sigma}_{j,z} \quad j=1, 2, \dots, M \text{ act on } |\mu\rangle$$

$$[\hat{\sigma}_{j,\alpha}, \hat{\sigma}_{m,\beta}] = 0 \text{ if } j \neq m$$

2D Ising
→ quasi-1D
spins chain

$$Z = \sum_{\{\sigma\}} e^{\beta J \sum_{r,c} \sigma_{r,c} \sigma_{r+1,c} + \beta J \sum_{r,c} \sigma_{r,c} \sigma_{r,c+1}}$$

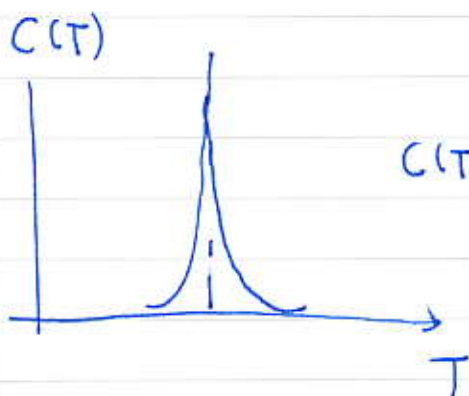
$$= \sum_{\{\sigma\}} \underbrace{e^{\beta J \sum_r \sigma_{r,1} \sigma_{r+1,1}}}_{\text{column 1}} \underbrace{e^{\beta J \sum_r \sigma_{r,1} \sigma_{r,2}}}_{\text{between column 1 \& 2}} \underbrace{e^{\beta J \sum_r \sigma_{r,2} \sigma_{r+1,2}}}_{\text{column 2}} \dots$$

$$V_1 = e^{K \sum_j \hat{\sigma}_{j,z} \hat{\sigma}_{j+1,z}} \quad V_2 = (2 \sinh 2K)^{\frac{M}{2}} e^{K^* \sum_{j=1}^M \hat{\sigma}_{j,x}}$$

free energy

$$\frac{\beta F(h=0, T)}{N} = -\ln(2 \cosh 2K) - \frac{1}{\pi} \int_0^{\frac{\pi}{2}} d\theta \ln \frac{1 + \sqrt{1 - q^2 \sin^2 \theta}}{2}$$

$$q = \frac{2 \sinh 2K}{\cosh^2 2K}$$



$$C(T) = -\frac{2k_B}{\pi} \left(\frac{2T}{K_0 T_c} \right)^2 \ln \left| 1 - \frac{T}{T_c} \right| + \text{const}$$

$$\frac{K_0 T_c}{J} = 2.269185 \quad \text{or} \quad \sinh(2K_c) = 1$$



due to C N Yang: 1952

$$\frac{M}{N} = \left(1 - \frac{1}{\sinh^4(2K)} \right)^{\frac{1}{8}} \quad T < T_c$$

$$\beta = \frac{1}{8}$$

$$\chi = \left. \frac{\partial m}{\partial h} \right|_{h=0} \sim |T - T_c|^{-\frac{7}{4}}$$

$$\gamma = \frac{7}{4}$$

Onsager
conjectured it
1949

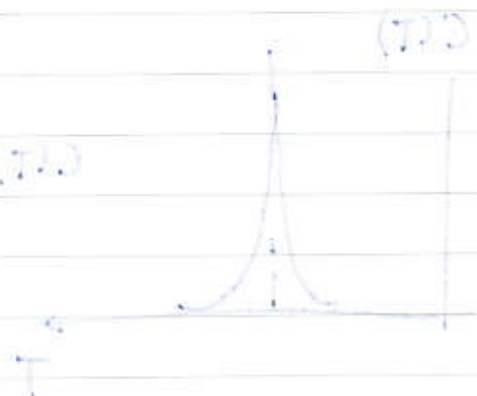
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$$\frac{\partial}{\partial T} \left(\frac{1}{T} \right) = -\frac{1}{T^2} \quad \frac{\partial}{\partial V} \left(\frac{1}{T} \right) = -\frac{1}{T^2} \frac{\partial T}{\partial V}$$

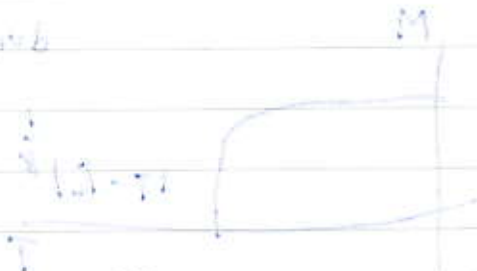
$$\frac{\partial \ln Q}{\partial T} = \frac{1}{T}$$

$$\frac{\partial}{\partial T} \left(\frac{1}{T} \right) = -\frac{1}{T^2} \quad \frac{\partial}{\partial V} \left(\frac{1}{T} \right) = -\frac{1}{T^2} \frac{\partial T}{\partial V}$$

$$\frac{\partial \ln Q}{\partial T} = \frac{1}{T}$$



$$\frac{\partial}{\partial T} \left(\frac{1}{T} \right) = -\frac{1}{T^2} \quad \frac{\partial}{\partial V} \left(\frac{1}{T} \right) = -\frac{1}{T^2} \frac{\partial T}{\partial V}$$



$$\frac{\partial}{\partial T} \left(\frac{1}{T} \right) = -\frac{1}{T^2}$$

$$T \lambda^{-\gamma} = 1 \quad \lambda = T^{\frac{1}{\gamma}}$$

$$Q(T, V, N) = \lambda^{3N} Q(T \lambda^{-\gamma}, V \lambda^{\frac{3}{\gamma}}, N)$$

$$Q(T, V, N) = T^{\frac{3N}{\gamma}} Q(1, V T^{-\frac{3}{\gamma}}, N)$$

$$P = -\frac{\partial F}{\partial V} = -\frac{\partial}{\partial V} (-k_B T \ln Q)$$

$$f(\lambda V T^{\frac{3}{\gamma}}, N) = \lambda^3 f(V T^{\frac{3}{\gamma}}, N)$$

$$= k_B T \frac{\partial}{\partial V} \left[\ln Q(1, V T^{-\frac{3}{\gamma}}, N) + \dots \right]$$

$$= k_B T \left[\left(\ln Q(1, V T^{-\frac{3}{\gamma}}, N) \right)' T^{-\frac{3}{\gamma}} \right]$$

because P is intensive no multi

$$= T^{1-\frac{3}{\gamma}} f(V T^{\frac{3}{\gamma}}, N) \rightarrow T^{1-\frac{3}{\gamma}} g\left(\frac{V}{N} T^{\frac{3}{\gamma}}\right)$$

Assign 3.

$$\textcircled{1} \quad \underline{2.3} \quad Z(T, V, N) = \frac{1}{h^{3N} N!} \int e^{-\beta H} d\vec{r}_1 d\vec{r}_2 \dots d\vec{r}_N d\vec{p}_1 \dots d\vec{p}_N$$

$$= \frac{1}{N!} \frac{1}{h^{3N}} \left[\int_{-\infty}^{\infty} e^{-\beta \frac{p^2}{2m}} dp \right]^{3N} \int e^{-\beta U} d\vec{r}_1 \dots d\vec{r}_N$$

$$= \frac{1}{N!} \frac{1}{h^{3N}} \left(\frac{2\pi m k_B T}{h^2} \right)^{\frac{3N}{2}} \cdot Q_N \quad \lambda^{-1} = \frac{\sqrt{2\pi m k_B T}}{h}$$

$$= \frac{1}{N!} (\lambda)^{-3N} Q_N$$

$$P = - \frac{\partial F}{\partial V} \quad T\text{-degms do not contribute}$$

$$Q(T, V, N) = \int_V d\vec{r}_1 \dots \int d\vec{r}_N e^{-\beta U(\vec{r}_1, \vec{r}_2, \dots, \vec{r}_N)}$$

$\int_0^L dx' \rightarrow \int_0^L dx$

$$r_n' = \lambda r_n$$

$$Q(T, V, N) \Rightarrow \lambda^N \int_{V\lambda^{-3}} d\vec{r}_1 \dots \int d\vec{r}_N e^{-\beta \lambda^{\sigma} U(\vec{r}_1, \dots, \vec{r}_N)}$$

$$\Rightarrow \lambda^N Q(T\lambda^{-\sigma}, V\lambda^{-3}, N)$$

$$\frac{V}{N} = v \quad Q(T, v) = \lambda^N Q(T\lambda^{-\sigma}, v\lambda^{-3}) \quad \lambda = v^{\frac{1}{3}}$$

true for any λ , choose $v\lambda^{-3} = 1$, ~~not~~ get or $T\lambda^{-\sigma} = 1$

$$Q(T, v) = v^N Q(Tv^{-\frac{\sigma}{3}}) = T^{-\frac{3N}{\sigma}} h(vT^{\frac{3}{\sigma}}) \quad \lambda = T^{\frac{1}{\sigma}}$$

$$P = - \frac{\partial (-k_B T) (\ln Q + \text{const})}{\partial V} = k_B T \frac{1}{Q} \frac{\partial Q}{\partial V} = k_B T \frac{1}{Q} \frac{\partial Q}{\partial v} \frac{1}{v} \cdot N$$

$$P = k_B T \cdot N \left[\frac{\partial \ln Q}{\partial V} \right]$$

$$= k_B T N \left[\frac{\partial}{\partial V} \left[N \ln V + \dots \right] \right]$$

$$= k_B T N \left[\frac{\partial}{\partial V} \ln \left(T^{-\frac{3N}{2}} h(V T^{\frac{3}{2}}) \right) \right]$$

$$= k_B T N \ln' \left(V T^{\frac{3}{2}} \right) T^{-\frac{3}{2}} = T^{-\frac{3}{2}} f \left(\frac{V}{N} T^{\frac{3}{2}} \right)$$

2.7. (a) $\omega_k = ck$

$$Z = \sum_{n_k=0,1,2,\dots} e^{-\beta \sum_k \omega_k n_k} = \prod_k \frac{1}{1 - e^{-\beta \hbar \omega_k}}$$

$$F = -k_B T \ln Z = -k_B T \ln \prod_k \frac{1}{1 - e^{-\beta \hbar \omega_k}}$$

$$= -k_B T \sum_k (-\ln(1 - e^{-\beta \hbar \omega_k}))$$

$$= k_B T \int \frac{(2V)}{(2\pi)^3} d\vec{k} \ln(1 - e^{-\beta \hbar \omega_k})$$

$$k = \frac{\omega}{c}$$

$$= k_B T \int \frac{(2V)}{(2\pi)^3} 4\pi k^2 dk \ln(1 - e^{-\beta \hbar \omega_k})$$

$$= \frac{V k_B T}{\pi^2 c^3} \int \omega^2 d\omega \ln(1 - e^{-\beta \hbar \omega})$$

$$= \frac{V k_B T}{\pi^2 c^3} \int_0^\infty d\omega \cdot \omega^2 \ln(1 - e^{-\beta \hbar \omega})$$

$$= \frac{V k_B T}{\pi^2 c^3} \left(-\frac{\pi^4}{45 (\beta \hbar)^3} \right) = -\frac{\pi^2 V}{45 c^3 \hbar^3 \beta^4}$$

(b)

$$P = -\frac{\partial F}{\partial V} = -\frac{k_B T}{\pi^2 c^3} \int_0^\infty d\omega \cdot \omega^2 \ln(1 - e^{-\beta \hbar \omega}) = \frac{\pi^2 177}{45 c^3 \hbar^3 \beta^4}$$

$$U = \frac{\partial \beta F}{\partial \beta} = \frac{\partial}{\partial \beta} \left(\frac{V}{\pi^2 c^3} \int_0^\infty d\omega \cdot \omega^2 \ln(1 - e^{-\beta \hbar \omega}) \right)$$

$$= \frac{V}{\pi^2 c^3} \int_0^\infty d\omega \cdot \omega^2 \frac{e^{-\beta \hbar \omega} \cdot \hbar \omega}{1 - e^{-\beta \hbar \omega}}$$

$$= \frac{V}{\pi^2 c^3} \int_0^\infty d\omega \cdot \omega^2 \cdot \frac{\hbar \omega}{e^{\beta \hbar \omega} - 1} = \frac{3 \pi^2 V}{45 c^3 \hbar^3 \beta^4}$$

$$PV = -\frac{k_B T V}{\pi^2 c^3} \int_0^\infty d\omega \cdot \omega^2 \ln(1 - e^{-\beta \hbar \omega})$$

$$= \frac{k_B T V}{\pi^2 c^3} \int_0^\infty \omega^2 \ln(1 - e^{-\beta \hbar \omega}) d\omega$$

$$= \frac{k_B T V}{\pi^2 c^3} \left[+ \frac{\omega^3}{3} \ln(1 - e^{-\beta \hbar \omega}) \Big|_0^\infty + \int_0^\infty \frac{\omega^3}{3} \frac{(-e^{-\beta \hbar \omega} (-\beta \hbar))}{1 - e^{-\beta \hbar \omega}} d\omega \right]$$

$$= \frac{1}{3} \frac{V}{\pi^2 c^3} \int_0^\infty \frac{\hbar \omega}{e^{\beta \hbar \omega} - 1} \omega^2 d\omega = \frac{1}{3} U$$

$\frac{1}{3}$ vs. $\frac{2}{3}$. disperse relation is different!

Date

Scaling of free energy

$$f(t, h) = \frac{F(T, h)}{N}$$

free energy per spin

$$t = \frac{T - T_c}{T_c}$$

 $t=0$ mean critical pt

$$f(t, h) = L^{-d} f(L^y t, L^x h) \quad \text{for any } L \quad \begin{array}{l} L \text{ for system size} \\ + \text{ regular (analytic piece)} \end{array}$$

Take $L^y t = 1$

$$L = t^{-\frac{1}{y}}$$

 d : dimension of systemuse "b" for L

$$f(t, h) = t^{+\frac{d}{y}} f(1, t^{-\frac{x}{y}} h)$$

here

 b : linear scaling factor

$$= t^{+\frac{d}{y}} \hat{f}(t^{-\frac{x}{y}} h)$$

$$m = -\frac{\partial f}{\partial h} = t^{+\frac{d}{y}} \hat{f}'(t^{-\frac{x}{y}} h) \cdot t^{-\frac{x}{y}} \Big|_{h=0} = t^{-\frac{-d+x}{y}} \hat{f}'(0) \propto t^{\beta}$$

$$\beta = -\frac{-d+x}{y} = \frac{-x+d}{y}$$

$$\chi = \frac{\partial m}{\partial h} = t^{-\frac{-d+x}{y}} \hat{f}''(t^{-\frac{x}{y}} h) t^{-\frac{x}{y}} \Big|_{h=0} = t^{-\frac{(-d)+2x}{y}} \propto t^{-\gamma}$$

$$\gamma = \frac{-d+2x}{y}$$

$$m = t^{\frac{d-x}{y}} \hat{f}'(t^{-\frac{x}{y}} h) = h^{\frac{d-x}{x}} \left(t^{-\frac{x}{y}} h \right)^{-\frac{y}{x} \left(\frac{d-x}{y} \right)} \hat{f}'(t^{-\frac{x}{y}} h) \Big|_{t=0} \propto h^{1/\delta}$$

$$\delta = \frac{x}{d-x}$$

$$\frac{\partial f}{\partial t} = -s$$

heat capacity

$$\frac{dU}{dt} = -s \frac{dt}{dt}$$

$$dF = -SdT - Mdh$$

heat capacity near T_c 181

$$U = F + TS$$

$$dU = \underbrace{T}_{dQ} dS - Mdh$$

$$C_h = \left. \frac{dQ}{dT} \right|_{h=0} = T \left. \frac{dS}{dT} \right|_{h=0} \approx T_c \left. \frac{\partial S}{\partial T} \right|_{h=0}$$

$$C_h = \frac{C_b}{N} = T_c \left(-\frac{\partial^2 f}{\partial t^2} \right) \bigg|_{h=0} \propto \frac{\partial^2}{\partial t^2} \left[t^{\frac{d}{y}-1} \hat{f}(t^{-\frac{x}{y}} h) \right] + t^{\frac{d}{y}} \hat{f}'(t^{-\frac{x}{y}} h) \left(-\frac{x}{y} t^{-\frac{x}{y}-1} h \right)$$

$$\propto t^{\frac{d}{y}-2} \hat{f}(0) \bigg|_{h=0} \propto t^{-\alpha}$$

$$\alpha = 2 - \frac{d}{y} \quad \text{by rescaling}$$

$$\alpha + 2\beta + \gamma = 2 - \frac{d}{y} + 2 \left(\frac{-x+d}{y} \right) + \frac{-d+2x}{y}$$

$$= 2 + \frac{-d-2x+2d-d+2x}{y} = 2$$

$$\boxed{\alpha + 2\beta + \gamma = 2} \quad \text{Rushbrooke's scaling law}$$

$$\beta(\delta-1) = \frac{d-x}{y} \left(\frac{x}{d-x} - 1 \right) = \frac{d-x}{y} \frac{x-d+x}{d-x} = \frac{2x-d}{y} = \gamma$$

$$\boxed{\beta(\delta-1) = \gamma} \quad \text{Widom's scaling law}$$

concept of Universality

critical exponents only depends on

- ① dimensionality d
- ② spin-dimension n
- ③ range of interaction

critical exponents

exponent	definition	2D Ising	3D Ising / fluid	mean-field
α	$C \sim t ^{-\alpha}$	0	0.12	?
β	$m \sim t ^\beta$	$\frac{1}{8}$	0.32	$\frac{1}{2}$
γ	$\chi \sim t ^{-\gamma}$	$\frac{7}{4}$	1.25	1
δ	$h \sim m^\delta$	15	5.0	3
$\nu = \frac{1}{\gamma}$	$\xi \sim t ^{-\nu}$	1	0.63	$\frac{1}{2}$
η	$G(r) \sim \frac{e^{-\frac{r}{\xi}}}{r^{d-2+\eta}}$	$\frac{1}{4}$	0.06	0

$$\chi = \int d^d r G(r) = \int \frac{e^{-\frac{r}{\xi}}}{r^{d-2+\eta}} r^{d-1} dr \approx \xi^{2-\eta} \approx |t|^{-\nu(2-\eta)} \sim |t|^{-\gamma}$$

$$\boxed{\gamma = \nu(2-\eta)} \quad \text{Fisher scaling law}$$

hyperscaling
(Josephson)

$$\boxed{d\nu = 2 - \alpha}$$

Finite-Size Scaling

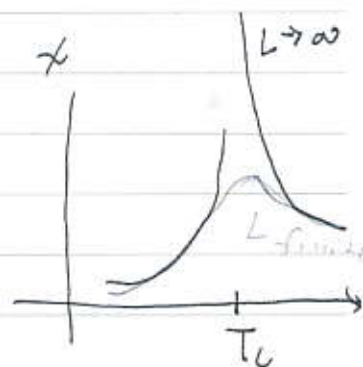
$$f(t, h, L) = b^{-d} f(b^x t, b^y h, b/L)$$

↑
size

finite-size rounding, e.g.



$$\chi(t, h=0, L) = L^{\gamma/\nu} \tilde{\chi}(t L^{1/\nu})$$



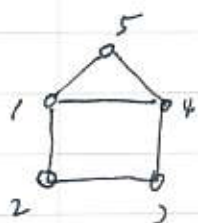
$$x = \tanh K$$

Renormalization group method.

$$N = 5$$

$$L = 6$$

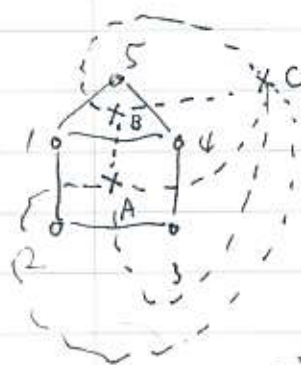
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$$Z = 2^5 (\cosh K)^6 \left(1 + \text{square} + \text{triangle} + \text{pentagon} \right)$$

$$= 2^5 (\cosh K)^6 (1 + x^4 + x^3 + x^5)$$

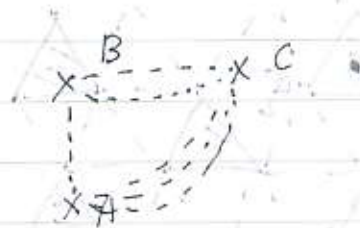
$$Z = \sum_{\{\sigma_i, \sigma_j\}} e^{K(\sigma_1\sigma_2 + \sigma_2\sigma_3 + \sigma_3\sigma_4 + \sigma_4\sigma_1 + \sigma_1\sigma_5 + \sigma_2\sigma_5)}$$



$$N^* = 3$$

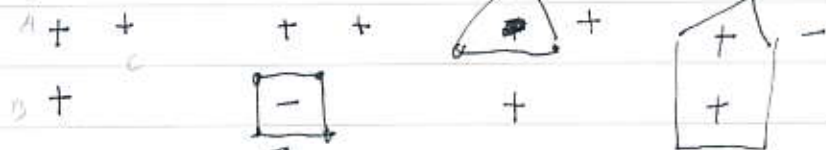
$$L^* = L = 6$$

dual lattice



$$Z^* = \sum_{\{\sigma_A, \sigma_B, \sigma_C\}} e^{K(\sigma_A\sigma_B + 2\sigma_B\sigma_C + 3\sigma_A\sigma_C)}$$

$$= 2 e^{K \cdot 6} (1 + e^{-8K} + e^{-6K} + e^{-10K})$$



$$2^3 = 8$$

Wall

boundary
connects
+ with -

We see p $V(r) = p^*(r) = 1$

$$r = 0, 4, 5$$

The low-temperature expansion coefficient on dual lattice p^* is the same as the high-temperature result of the original lattice

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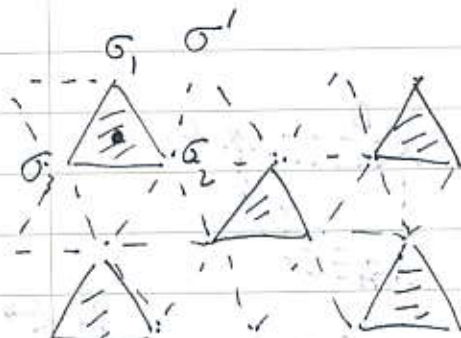
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$$(x^2 + x + 1)^3 (x^2 + x + 1)^3 = 2$$

$$(x^2 + x + 1)^3 (x^2 + x + 1)^3 =$$

$$(x^2 + x + 1)^3 (x^2 + x + 1)^3 =$$

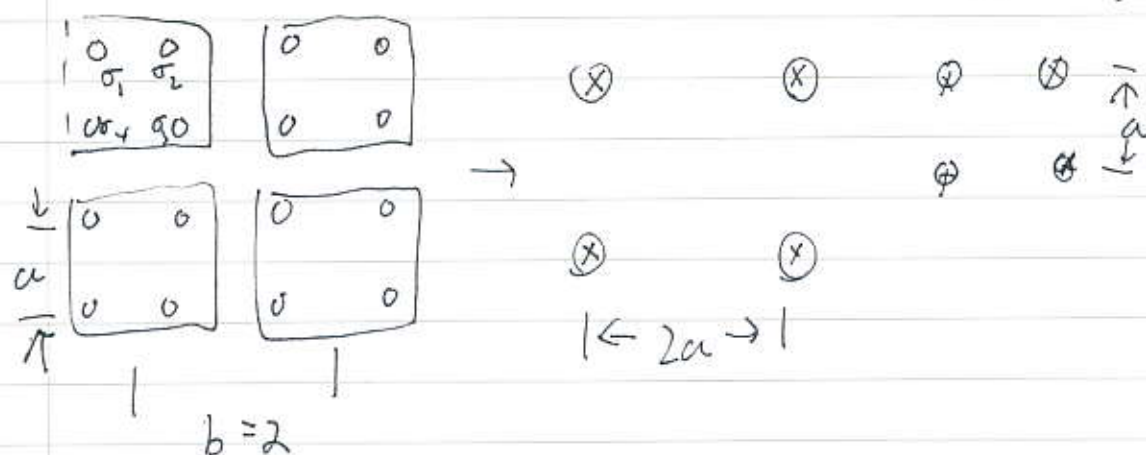
Block spin transfer on triangular lattice



$$P(\sigma', \sigma) = \prod \frac{1}{2} \left[1 + \sigma' (\sigma_1 + \sigma_2 + \sigma_3 - \sigma_1 \sigma_2 \sigma_3) \right]^{\frac{1}{2}}$$

RG method

Kadanoff block spin transformation

Scaling picture
scaling invariant
at T_c 

new spin $\sigma_1' = \text{majority of } (\sigma_1, \sigma_2, \sigma_3, \sigma_4)$
 or sign of $\sum_{i=1}^4 \sigma_i$.

$$H = \sum_{\alpha} K_{\alpha} S_{\alpha}$$

$$S_{\alpha} = \prod_{i \in I_{\alpha}} \sigma_i$$

I_{α} empty set
 nn,
 site site

$$= K_0 + K_1 \sum_i \sigma_i + K_2 \sum_{\langle i,j \rangle} \sigma_i \sigma_j + K_3 \sum_{\langle i,j,k \rangle} \sigma_i \sigma_j \sigma_k + \dots$$

$$Z_N = \sum_{\{\sigma\}} e^{-H(\sigma)}$$

$$Z_{N'} = \sum_{\{\sigma'\}} e^{-H'(\sigma')}$$

renormalize H'

$$Z_N = Z_{N'}$$

$$e^{-H'(\sigma')} = \sum_{\{\sigma\}} P(\sigma', \sigma) e^{-H(\sigma)}$$

Block spin
 decimation

$$\sum_{\sigma'} P(\sigma', \sigma) = 1 \quad P(\sigma', \sigma) \geq 0$$

RG transfn $H(\sigma) \rightarrow H'(\sigma')$ or $\{K_i\} \rightarrow \{K'_i\}$

$$\vec{K}' = f(\vec{K})$$

$$\vec{K}'' = f(\vec{K}') \dots$$

$$\vec{K}_{n+1}^{(h)} = f(\vec{K}_n^{(h)})$$

$n \rightarrow \infty$

fixed point.

$$\vec{K}^* = f(\vec{K}^*)$$

Date

Expanding around fixed pt.

$$\delta \vec{K}_{n+1}^{(h)} = \vec{K}_{n+1}^{(h)} - \vec{K}^* = f(\vec{K}_n^{(h)}) - f(\vec{K}^*) \\ = W(\vec{K}_n^{(h)} - \vec{K}^*)$$

$$\rightarrow \delta \vec{K}_{n+1}^{(h)} = W \delta \vec{K}_n^{(h)} \quad \leftarrow \text{linearized transform}$$

Let $\phi W = \lambda \phi$ left eigen vector of W

$$v^{(h)} = \sum_{\alpha} \phi_{\alpha} (\delta K^{(h)})_{\alpha}$$

$$v^{(h+1)} = \sum_{\alpha} \phi_{\alpha} (\delta K^{(h+1)})_{\alpha} = \sum_{\alpha} \underbrace{\phi_{\alpha} W_{\alpha\beta}}_{\lambda \phi_{\alpha}} \delta K_{\beta}^{(h)} = \lambda \sum_{\beta} \phi_{\beta} \delta K_{\beta}^{(h)} \\ = \lambda v^{(h)}$$

$\rightarrow$ new scaling fixed

original

new variable

$\lambda > 1$ relevant

$\lambda < 0$ irrelevant

$\lambda = 0$ marginal

$$\{K\} \rightarrow \{v\}$$

$\lambda = b^y \leftarrow$ critical exponent

free magnetic

In symbol cases

$$v_1 = t = \frac{T - T_c}{T}$$

$$v_2 = h$$

RG transform

$$t' = \lambda_T t = b^{Y_T} t$$

$$Y_T = y$$

$$h' = \lambda_H h = b^{Y_H} h$$

$$Y_H = x$$

$$Z_N(K) = Z_{N'}(K')$$

$$\frac{N'}{N} = b^{-d}$$

$$N f(\{v\}) = N' f(\{v'\}) \Rightarrow f(t, h) = b^{-d} f(b^{Y_T} t, b^{Y_H} h)$$

At fixed point $\beta = 0$, or ∞

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$$\beta(t, h) = b \beta(b^Y t, b^X h)$$

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n iteration $\beta(0,0) = b^n \beta(q,0)$

Field-theoretic approach

Ginsburg-Landau - Wilson model

ϕ^4 theory

$$H(\phi) = \int dx \left\{ \frac{1}{2} (\nabla \phi)^2 + \frac{1}{2} r_0 \phi^2 + \frac{1}{4} u_0 \phi^4 \right\} \quad \phi(x)$$

Universality from the RG point of view.

RG results (ϵ -expansion)

$$\epsilon = 4-d$$

$$\beta = \frac{1}{2} - \frac{3\epsilon}{2(n+8)} + \frac{\epsilon^2 (n+2)(2n+1)}{2(n+8)^3} + \dots$$

upto ϵ^5
are presently known

$$\nu = \frac{1}{2} + \frac{\epsilon (n+2)}{4(n+8)} + \frac{\epsilon^2 (n+2)}{8(n+8)^3} (n^2 + 23n + 60) + \dots$$

K Wilson found ϵ^2 order

n dimensions of the spin

$$n=1 \quad \text{Ising}$$

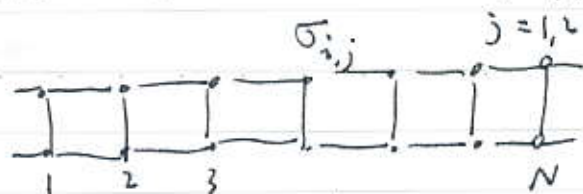
$$n=2 \quad \text{XY model}$$

$$n=3 \quad \text{Heisenberg model}$$

Tutorial 4

 $i=1,2 \dots N$

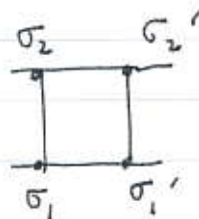
4.



Date

$$Z = \sum_{\{\sigma\}} e^{K \sum_{i=1}^N (\sigma_{i,1} \sigma_{i+1,1} + \sigma_{i,2} \sigma_{i+1,2} + \frac{1}{2} \sigma_{i,1} \sigma_{i,2} + \frac{1}{2} \sigma_{i+1,1} \sigma_{i+1,2})}$$

$$= \sum_{\{\sigma\}} \prod_i e^{K (\sigma_{i,1} \sigma_{i+1,1} + \sigma_{i,2} \sigma_{i+1,2} + \frac{1}{2} \sigma_{i,1} \sigma_{i,2} + \frac{1}{2} \sigma_{i+1,1} \sigma_{i+1,2})}$$



$$V(\sigma_1 \sigma_2; \sigma'_1 \sigma'_2)$$

$$= e^{K (\sigma_1 \sigma'_1 + \sigma_2 \sigma'_2 + \frac{1}{2} \sigma_1 \sigma_2 + \frac{1}{2} \sigma'_1 \sigma'_2)}$$

$$V = \begin{matrix} \begin{matrix} \uparrow \uparrow \\ \sigma_1 \sigma_2 \end{matrix} & \begin{matrix} ++ & +- & -+ & -- \end{matrix} \\ \begin{matrix} ++ \\ +- \\ -+ \\ -- \end{matrix} & \begin{bmatrix} e^{3K} & 1 & 1 & e^{-K} \\ 1 & e^K & e^{3K} & 1 \\ 1 & e^{3K} & e^K & 1 \\ e^{-K} & 1 & 1 & e^{3K} \end{bmatrix} \end{matrix}$$

$$\rightarrow \begin{matrix} \begin{matrix} +- & +- \\ ++ & +- \\ -+ & +- \\ ++ & +- \end{matrix} & \begin{matrix} +- & +- \\ +- & +- \\ -+ & +- \\ -+ & +- \end{matrix} \end{matrix}$$

$$1+1 + \frac{1}{2}(-1-1)$$

largest eigen value

$$\lambda_{\max} = \frac{1}{2} e^{-4K} (e^K + e^{3K} + e^{5K} + e^{7K})$$

$$+ e^K (1 + e^{2K}) \sqrt{1 - 4e^K + 10e^{4K} - 4e^{6K} + e^{8K}}$$

$$Z = \text{Tr}(V^N) = \lambda_1^N + \lambda_2^N + \lambda_3^N + \lambda_4^N$$

$$N \rightarrow \infty \quad F = -N k_B T \ln \lambda_{\max}$$

eigenvalues

$$\det(V - \lambda I) = 0$$

$$e^K = x$$

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$$\det \begin{vmatrix} e^{3K} - \lambda & 1 & 1 & e^{-K} \\ 1 & e^K - \lambda & e^{-3K} & 1 \\ 1 & e^{-3K} & e^K - \lambda & 1 \\ e^{-K} & 1 & 1 & e^{3K} - \lambda \end{vmatrix} = 0$$

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$$\begin{vmatrix} x^3 - \lambda & 1 & 1 & x^{-1} \\ 1 & x - \lambda & x^{-3} & 1 \\ 1 & x^{-3} & x - \lambda & 1 \\ x^{-1} & 1 & 1 & x^3 - \lambda \end{vmatrix} = 0$$

 $|A_{ij}|$ minor

$$A_{ij} = (-1)^{i+j} |A_{ij}|$$

← cofactor

Laplace expansion

1st row

$$|A| = \sum_j a_{ij} A_{ij}$$

for fixed i

$$= (x^3 - \lambda) \begin{vmatrix} x - \lambda & x^{-3} & 1 \\ x^{-3} & x - \lambda & 1 \\ 1 & 1 & x^3 - \lambda \end{vmatrix} - \begin{vmatrix} 1 & x^{-3} & 1 \\ 1 & x - \lambda & 1 \\ x^{-1} & 1 & x^3 - \lambda \end{vmatrix} + \begin{vmatrix} 1 & x - \lambda & 1 \\ 1 & x^{-3} & 1 \\ x^{-1} & 1 & x^3 - \lambda \end{vmatrix}$$

$$- x^{-1} \begin{vmatrix} 1 & x - \lambda & x^{-3} \\ 1 & x^{-3} & x - \lambda \\ x^{-1} & 1 & 1 \end{vmatrix}$$

$$= (x^3 - \lambda) \left[(x - \lambda)^2 (x^3 - \lambda) + 2x^{-3} - (x - \lambda) - x^{-6} (x^3 - \lambda) - (x - \lambda) \right]$$

$$- \left[(x - \lambda) (x^3 - \lambda) + x^{-3} (x^3 - \lambda) + x^{-1} (x - \lambda) - \dots \right]$$

$$= e^{-8K} (-1 + e^{4K} - e^K \lambda) (-1 + e^{4K} - e^{3K} \lambda) x (1 - 2e^{4K} + e^{8K} - (e^K + e^{3K} + e^{5K} + e^{7K}) \lambda + e^{4K} \lambda^2)$$

$$\lambda_1 = e^{3K} - e^{-K}$$

$$\lambda_2 = e^K - e^{-3K}$$

$$\lambda_{3,4} = \frac{1}{2e^{4K}} \left[e^K + e^{3K} + e^{5K} + e^{7K} \pm \sqrt{(e^K + e^{3K} + e^{5K} + e^{7K})^2 - 4e^{4K}(1 - 2e^{4K} + e^{8K})} \right]$$

as $K \rightarrow \infty$

$$\lambda_1 \propto e^{3K}$$

$$\lambda_2 \propto e^K$$

$$\lambda_{3,4} \propto e^{1K}$$

5. (a) $\langle M \rangle = \left\langle \sum_{i=1}^N \sigma_i \right\rangle = \frac{1}{Z} \sum_{\{\sigma\}} \left(\sum_{i=1}^N \sigma_i \right) e^{K \sum_{\langle ij \rangle} \sigma_i \sigma_j + \beta h \sum_i \sigma_i}$

$$\chi = \left. \frac{\partial \langle M \rangle}{\partial h} \right|_{h=0} = \frac{\partial}{\partial h} \left(\frac{1}{Z} \sum_{\{\sigma\}} M(\sigma) e^{K \sum_{\langle ij \rangle} \sigma_i \sigma_j + \beta h M(\sigma)} \right)$$

$$= \frac{1}{Z} \sum_{\{\sigma\}} M(\sigma) \frac{\partial}{\partial h} e^{K \sum_{\langle ij \rangle} \sigma_i \sigma_j + \beta h M(\sigma)} - \frac{\sum_{\{\sigma\}} M(\sigma) e^{K \sum_{\langle ij \rangle} \sigma_i \sigma_j + \beta h M(\sigma)}}{Z^2} \frac{\partial Z}{\partial h}$$

$$= \frac{1}{Z} \sum_{\{\sigma\}} M(\sigma) \cdot \beta M(\sigma) e^{K \sum_{\langle ij \rangle} \sigma_i \sigma_j + \beta h M(\sigma)} - \frac{\sum_{\{\sigma\}} M(\sigma) e^{K \sum_{\langle ij \rangle} \sigma_i \sigma_j + \beta h M(\sigma)}}{Z} \frac{1}{Z} \frac{\partial Z}{\partial h}$$

$$= \beta \langle M(\sigma)^2 \rangle - \langle M \rangle \frac{1}{Z} \frac{\partial Z}{\partial h}$$

$$\frac{1}{Z} \frac{\partial Z}{\partial h} = \frac{1}{Z} \sum_{\{\sigma\}} \frac{\partial}{\partial h} e^{K \sum_{\langle ij \rangle} \sigma_i \sigma_j + \beta h \sum_i \sigma_i} = \frac{1}{Z} \sum_{\{\sigma\}} (\beta M(\sigma)) e^{K \sum_{\langle ij \rangle} \sigma_i \sigma_j + \beta h \sum_i \sigma_i}$$

$$= \beta \langle M \rangle$$

so $\chi = \beta \left[\langle M(\sigma)^2 \rangle - \langle M(\sigma) \rangle^2 \right] \quad \beta = \frac{1}{k_B T}$

(b) $\langle M \rangle = \frac{1}{Z} \sum_{\{\sigma\}} \sum_{i=1}^N e^{K \sum_{\langle ij \rangle} \sigma_i \sigma_j}$ set $h=0$

$= \frac{1}{Z} \sum_{\{\sigma\}} \left(\sum_{i=1}^N \sigma_i \right) (\cosh^L K) \prod_{\langle ij \rangle} (1 + \sigma_i \sigma_j \tanh K) = 0$

there are always odd # of spins at each site

$\langle M^2 \rangle = \frac{1}{Z} \sum_{\{\sigma\}} \left(\sum_{i=1}^N \sigma_i \right)^2 (\cosh^L K) \prod_{\langle ij \rangle} (1 + \sigma_i \sigma_j \tanh K)$

$Z = \sum_{\{\sigma\}} (\cosh^L K) \prod_{\langle ij \rangle} (1 + \sigma_i \sigma_j \tanh K)$

$= \frac{\sum_{\{\sigma\}} \sum_{k,l} \sigma_k \sigma_l \prod_{\langle ij \rangle} (1 + \sigma_i \sigma_j x)}{\sum_{\{\sigma\}} \prod_{\langle ij \rangle} (1 + \sigma_i \sigma_j x)}$

$\sum_{\{\sigma\}} \prod_{\langle ij \rangle} (1 + \sigma_i \sigma_j x)$

$k=l$
 $= \sum_{k=1}^N \sigma_k^2 \sum_{\{\sigma\}} \prod_{\langle ij \rangle} (1 + \sigma_i \sigma_j x) / \sum_{\{\sigma\}} \prod_{\langle ij \rangle} (1 + \sigma_i \sigma_j x)$

$\frac{k-l}{x}$

$= N +$

k, l are nearest
neigh

of bonds

$\sum_{\{\sigma\}} \prod_{\langle ij \rangle} (1 + \sigma_i \sigma_j x) = 1 + \dots x^4$

$= N + x \cdot N \cdot d \cdot 2 + x^2 \cdot N \cdot \frac{2d(2d-1)}{2} + \dots$

$\chi = \frac{N}{K_0 T} (1 + 2d \cdot x + \dots)$

K can be on 1st or
for each bond of $\prod_{\langle ij \rangle}$ each site
there are two spins

$\chi = \frac{N}{K_0 T} (1 + 2d \cdot x + 2d(2d-1) x^2 + \dots)$

$= \frac{N}{K_0 T} (1 + z \cdot x + z(z-1) x^2 + \dots)$ $z=2d$ is coordination number

Q7 $E = -\sum \mu_B H \cos \theta$

Date

(a) $Z = z^N$ $z = \int e^{-\beta E} \sin \theta d\theta d\phi$

$$= 2\pi \int_0^\pi e^{\beta \mu_B H \cos \theta} \sin \theta d\theta$$

$$= 2\pi \int_0^\pi e^{\beta \mu_B H \cos \theta} [-d \cos \theta]$$

$$= 2\pi \left[-\int_1^{-1} e^{\beta \mu_B H x} dx \right] = \frac{2\pi (-1)}{\beta \mu_B H} \left[e^{\beta \mu_B H x} \right]_1^{-1}$$

$$= \frac{2\pi (-1)}{\beta \mu_B H} [e^{-\beta \mu_B H} - e^{\beta \mu_B H}] = \frac{4\pi}{\beta \mu_B H} \sinh(\beta \mu_B H)$$

$\cos \theta = x$
 $\sinh x = x + \frac{1}{6}x^3 + \dots$

(b) $F = -k_B T \ln Z^N$

$$= -N k_B T \ln \frac{2\pi}{\beta \mu_B H} \sinh(\beta \mu_B H)$$

$$= -N k_B T \left[\ln(2\pi) + \ln \left(1 + \frac{1}{6} (\beta \mu_B H)^2 + \dots \right) \right]$$

$$= -N k_B T \left(\ln(2\pi) + \frac{1}{6} (\beta \mu_B H)^2 + \dots \right)$$

$$= -N k_B T \cdot C - \frac{1}{6} N \frac{(\mu_B H)^2}{k_B T} + \dots$$

$$S = -\frac{\partial F}{\partial T} = N k_B C + \left(-\frac{1}{6}\right) N \frac{(\mu_B H)^2}{k_B T^2}$$

$$= N k_B \left(C - \frac{1}{6} \left(\frac{\mu_B H}{k_B T} \right)^2 \right)$$

(c) $\Delta Q (B \rightarrow C) = 0$ $\Delta Q = T \Delta S$ $dQ = T dS$

$$\Delta Q (A \rightarrow B) = T_1 (\Delta S) = T_1 \left(-\frac{N \mu_B^2}{6 k_B T_1^2} (H_2^2 - H_1^2) \right)$$

reversed

$$\Delta Q = \frac{1}{6} \frac{N \mu_B^2}{k_B T_1} (H_2^2 - H_1^2) > 0$$

$$D \text{ (NaCl in water)} \sim 1.2 \times 10^{-10} \text{ m}^2/\text{s}$$

$$\langle x^2 \rangle \sim Dt$$

$$\begin{array}{l} \text{gas transport } 10^{-6} \\ \text{solid in solid } 10^{-12} \end{array} \left\{ \text{m}^2/\text{s} \right\}$$

Date

$$g \left\{ \begin{array}{l} \text{gas} \\ \text{solid} \end{array} \right\} = \frac{1}{2} \quad M_s = \frac{1}{2} \quad (a)$$

$$R. \text{ Kubo, M. Toda, N. Hashi}$$

NS = "statistical physics II"

$$R. \text{ Zwanzig, "Nonequilibrium Statistical Mechanics"} \\ \text{NS} =$$

$$N. \text{ Potier, "Nonequilibrium Statistical Physics"} \\ \text{NS} =$$

$$J(u)_{\text{NS}} = \left[\frac{1}{2} \left(\frac{1}{2} \right) \right] \text{NS} =$$

$$\left(\frac{1}{2} \right) \text{NS} = \left[\frac{1}{2} \left(\frac{1}{2} \right) \right] \text{NS} =$$

$$\frac{1}{2} + \frac{1}{2} = 1$$

$$\frac{1}{2} \left(\frac{1}{2} \right) = \frac{1}{2} \quad (b)$$

$$\left(\frac{1}{2} \right) \text{NS} = \frac{1}{2} \left(\frac{1}{2} \right) \text{NS} =$$

$$\left(\frac{1}{2} \right) \text{NS} = \frac{1}{2} \left(\frac{1}{2} \right) \text{NS} =$$

$$\left(\frac{1}{2} \right) \text{NS} = \frac{1}{2} \left(\frac{1}{2} \right) \text{NS} =$$

Einstein 1905

Smoluchowski

$$\frac{1}{2} \left(\frac{1}{2} \right) \text{NS} = \frac{1}{2} \left(\frac{1}{2} \right) \text{NS} =$$

$$\frac{1}{2} \left(\frac{1}{2} \right) \text{NS} = \frac{1}{2} \left(\frac{1}{2} \right) \text{NS} =$$

$$\left(\frac{1}{2} \right) \text{NS} = \frac{1}{2} \left(\frac{1}{2} \right) \text{NS} =$$

$$2\delta T = \delta b$$

$$2\delta T = \delta \Delta$$

$$\delta \left(\frac{1}{2} \right) \delta \Delta = \delta \Delta \quad (c)$$

$$\left(\frac{1}{2} \right) \text{NS} = \frac{1}{2} \left(\frac{1}{2} \right) \text{NS} =$$

$$\left(\frac{1}{2} \right) \text{NS} = \frac{1}{2} \left(\frac{1}{2} \right) \text{NS} =$$

Brownian motion

① pollen floating in water making irregular motion (1827) 21 March 203

A large number of Brownian

② mirror hits by air molecules model an suspension fiber

Fick's law

③ noises in electric circuits

$$j = -D \frac{\partial n}{\partial x}$$

$n(x)$

1D (3D is similar)

$$\frac{\partial n}{\partial t} + \nabla \cdot j = 0$$

continuity / conservation of particles

→ diffusion Eq.

$$\frac{\partial n}{\partial t} = D \frac{\partial^2 n}{\partial x^2}$$

solution

transit from (x_0, t_0) to x, t

$$n(x, t) = \int n(x_0, t_0) P(x_0, t_0 | x, t) dx_0$$

initial density

for open b.c.

$$P(x_0, t_0 | x, t) = \frac{1}{\sqrt{4\pi D(t-t_0)}} e^{-\frac{(x-x_0)^2}{4D(t-t_0)}}$$

$$P(x_0, t_0 | x_2, t_2) = \int P(x_0, t_0 | x_1, t_1) dx_1 P(x_1, t_1 | x_2, t_2)$$

→ Markovian stochastic process

$$\frac{\partial P(x, t)}{\partial t} - D \frac{\partial^2 P(x, t)}{\partial x^2} = \delta(t-t_0) \delta(x-x_0)$$

$$P(x, t) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \tilde{P}(k, \omega) e^{i(\omega t - kx)} dk d\omega$$

$$\int \int \tilde{P}(k, \omega) [(i\omega) - D(-ik)^2] e^{i(\omega t - kx)} dk d\omega = \int \int \frac{1}{(2\pi)^2} e^{i(\omega t - kx)} e^{-i(\omega t_0 - kx_0)} dk d\omega$$

$$\tilde{P}(k, \omega) = \frac{1}{(2\pi)^2} \frac{e^{-i(\omega t_0 - kx_0)}}{-DK^2 + i\omega}$$

$$P(x,t) = \int_{-\infty}^{\infty} dk \int_{-\infty}^{\infty} \frac{dw}{(2\pi)^2} \frac{e^{i[w(t-t_0) + k(x-x_0)]}}{-Dk^2 + iw} = ? \quad 205$$

$$\frac{\partial}{\partial t} \left\{ \frac{1}{\sqrt{4\pi Dt}} e^{-\frac{x^2}{4Dt}} \right\} = \frac{1}{\sqrt{4\pi D}} \frac{(-\frac{1}{2})}{t^{\frac{3}{2}}} e^{-\frac{x^2}{4Dt}} + \frac{1}{\sqrt{4\pi Dt}} e^{-\frac{x^2}{4Dt}} \left[\frac{x^2}{4Dt^2} \right]$$

$$= \frac{1}{\sqrt{4\pi Dt}} \left[-\frac{1}{2t} + \frac{x^2}{4Dt^2} \right] e^{-\frac{x^2}{4Dt}}$$

$$\frac{\partial}{\partial x} \left\{ \frac{1}{\sqrt{4\pi Dt}} e^{-\frac{x^2}{4Dt}} \right\} = \frac{e^{-\frac{x^2}{4Dt}}}{\sqrt{4\pi Dt}} \left[-2 \frac{x}{4Dt} \right]$$

$$\frac{\partial^2}{\partial x^2} \left\{ \frac{1}{\sqrt{4\pi Dt}} e^{-\frac{x^2}{4Dt}} \right\} = \frac{e^{-\frac{x^2}{4Dt}}}{\sqrt{4\pi Dt}} \left[\frac{x^2}{4D^2 t^2} - \frac{1}{2Dt} \right] = \frac{\partial}{\partial t} \left\{ \frac{1}{\sqrt{4\pi Dt}} e^{-\frac{x^2}{4Dt}} \right\} \cdot \frac{1}{D}$$

$$D \frac{\partial^2}{\partial x^2} P$$

$$\int_{-\infty}^{\infty} P(x,t) dx = 1$$

$$\text{set } x_0 = t_0 = 0$$

$$P(x,t) = \frac{1}{\sqrt{4\pi Dt}} e^{-\frac{x^2}{4Dt}}$$

prob. that particle is at locatiⁿ x at time t , $\langle x(t) \rangle = 0$

$$\langle x(t)^2 \rangle = \int_{-\infty}^{\infty} \frac{1}{\sqrt{4\pi Dt}} e^{-\frac{x^2}{4Dt}} x^2 dx$$

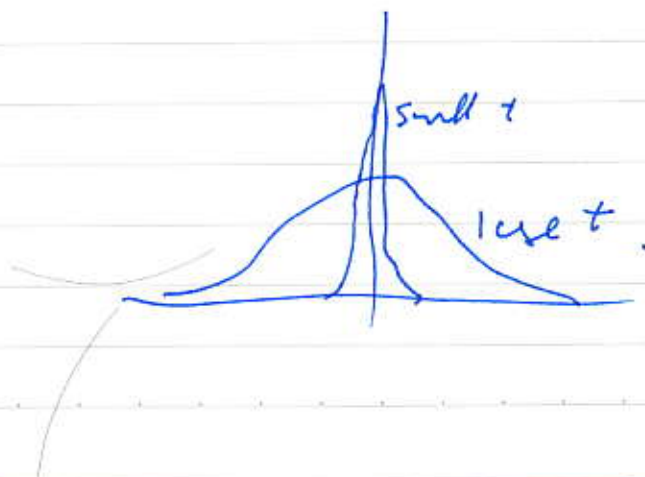
$$\frac{x^2}{2Dt} = y^2$$

$$x = \sqrt{2Dt} y$$

$$= \int_{-\infty}^{\infty} \frac{1}{\sqrt{4\pi Dt}} e^{-\frac{y^2}{2}} 2Dt y^2 \sqrt{2Dt} dy$$

$$= 2Dt \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} y^2 e^{-\frac{y^2}{2}} dy = 2Dt$$

$$\text{mean-square displacement} = 2Dt$$



A Brownian particle can be modeled as random walk 207

$$x(t) = \sum_{i=1, 2, \dots}^t \xi_i a$$

each step move to left or right with equal probability by a In time Δt .

$$\langle x(t) \rangle = 0 \quad \langle \xi_i \rangle = 0$$

$$\text{total \# of } N = \frac{t}{\Delta t}$$

ξ_i random value with prob

$$\begin{matrix} +\frac{1}{2} & +1 \\ +\frac{1}{2} & -1 \end{matrix}$$

$$\langle \xi_i \xi_j \rangle = \delta_{ij}$$

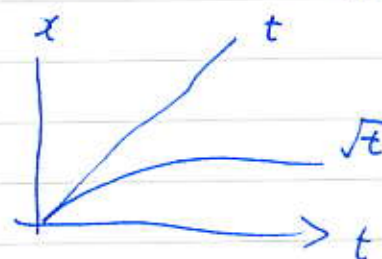
$$\langle x(t)^2 \rangle = \sum_{i=1}^N \sum_{j=1}^N \langle \xi_i \xi_j \rangle a^2 = N a^2 = t \frac{a^2}{\Delta t} = 2Dt$$

we can identify $D = \frac{1}{2} \frac{a^2}{\Delta t}$

$$[D] = L^2 T^{-1} \text{ m}^2 \text{ sec}^{-1}$$

ballistic motion $x \sim vt$

diffusive motion $x \sim \sqrt{Dt}$



$$\langle x^2 \rangle = \langle x^2 \rangle$$

$$\langle x^2 \rangle = \frac{\int_{-\infty}^{\infty} x^2 n(x) dx}{\int_{-\infty}^{\infty} n(x) dx} = \frac{\int_{-\infty}^{\infty} x^2 n(x) dx}{N}$$

$$\langle x^2 \rangle = \frac{\int_{-\infty}^{\infty} x^2 n(x) dx}{N}$$

$$\frac{\partial \langle x^2 \rangle}{\partial t} = D \frac{\partial^2 \langle x^2 \rangle}{\partial x^2}$$



from diffusion equation to $\langle x^2 \rangle = 2Dt$

$$\frac{\partial n}{\partial t} = D \frac{\partial^2 n}{\partial x^2}$$

$n \rightarrow$ density of particles
or prob of a single particle

n is treated as probability

$$\int n dx = 1$$

$$\langle x^2 \rangle = \int_{-\infty}^{\infty} x^2 n(x) dx$$

area under curve

$$x^2 \frac{\partial n}{\partial t} = D x^2 \frac{\partial^2 n}{\partial x^2}$$

$$n(\pm\infty) = 0$$

integrate:

$$\frac{\partial \langle x^2 \rangle}{\partial t} = D \int dx x^2 \frac{\partial^2 n}{\partial x^2}$$

$$= D \left[2x^2 \frac{\partial n}{\partial x} \right]_{-\infty}^{\infty} - \int_{-\infty}^{\infty} \frac{\partial x^2}{\partial x} \frac{\partial n}{\partial x} dx$$

$$= D \int_{-\infty}^{\infty} 2x \frac{\partial n}{\partial x} dx = D \left[2x n \right]_{-\infty}^{\infty} + \int_{-\infty}^{\infty} 2n dx$$

$$= 2D$$

$$\langle x^2 \rangle = 2Dt$$

also $\langle x^2 \rangle = 0$ at $t=0$

Langevin Eq

Date 22 March 07

and solution $\left\{ \begin{array}{l} \langle v \rangle \\ \langle v^2 \rangle \\ \langle x^2 \rangle \end{array} \right. \quad \langle v(0) v(t) \rangle$

fluctuation dissipation theorem

work in 1D (high dimensions is similar)

 $\langle \rangle$ average over what?

Stochastic differential Eq.

$$m \frac{dv}{dt} = -m\gamma v + R(t)$$

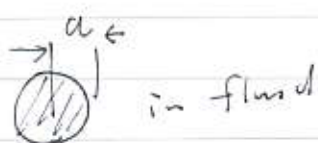
$$\langle R(t) \rangle = 0$$

$\uparrow$ friction $\uparrow$ random noise

$$\langle R(t) R(t') \rangle$$

$$= C \delta(t-t')$$

white noise



in fluid

$$m\gamma = 6\pi a \eta$$

$$F = -6\pi a \eta v \quad (\text{Stokes law})$$

η viscosity of fluid

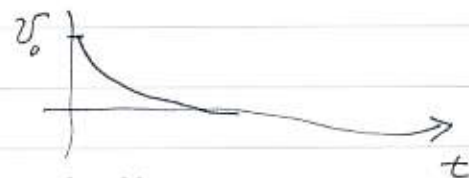
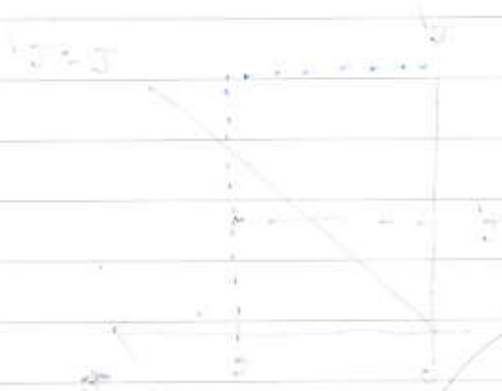
C is Not arbitrary if the system is in thermal equilibrium

 $\langle v \rangle$

$$m \frac{d\langle v \rangle}{dt} = -m\gamma \langle v \rangle$$

$$\frac{d\langle v \rangle}{dt} = -\gamma \langle v \rangle$$

$$\rightarrow \langle v \rangle = v_0 e^{-\gamma t}$$



average velocity decay to zero for any initial velocity

Let $v = A(t) e^{-\gamma t}$ ← method of variation of constant

$$\frac{dv}{dt} = \dot{A}(t) e^{-\gamma t} - \gamma A(t) e^{-\gamma t} = -\gamma A(t) e^{-\gamma t} + \frac{R(t)}{m}$$

$$\dot{A}(t) = \frac{1}{m} e^{\gamma t} R(t)$$

$$A(t) = \int_0^t \frac{1}{m} e^{\gamma t'} R(t') dt' + A(0)$$

$$V(t) = \left[\frac{1}{m} \int_0^t e^{\gamma t'} R(t') dt' + A(0) \right] e^{-\gamma t}$$

$$= \frac{1}{m} \int_0^t e^{-\gamma(t-t')} R(t') dt' + V(0) e^{-\gamma t}$$

again $\langle V(t) \rangle = V(0) e^{-\gamma t}$ as expected $\langle R \rangle = 0$

$$\langle V(t) V(t') \rangle = \left\langle \frac{1}{m} \int_0^t e^{-\gamma(t-\tau)} R(\tau) d\tau \frac{1}{m} \int_0^{t'} e^{-\gamma(t'-\tau')} R(\tau') d\tau' \right\rangle$$

cross terms vanish because

$$\langle R(t) V_0 \rangle = 0$$

$$= \frac{1}{m^2} \int_0^t d\tau \int_0^{t'} d\tau' e^{-\gamma(t-\tau + t' - \tau')} \underbrace{\langle R(\tau) R(\tau') \rangle}_{C \delta(\tau - \tau')}$$

$$= \frac{1}{m^2} \int_0^t d\tau \int_0^{t'} d\tau' e^{-\gamma(t-\tau + t' - \tau')} C \delta(\tau - \tau') \quad \text{assuming } t' > t$$

$$= \frac{C}{m^2} \int_0^t d\tau e^{-\gamma(t+t'-2\tau)} = \frac{C}{m^2} e^{-\gamma(t+t')} \int_0^t e^{2\gamma\tau} d\tau$$

$$= \frac{C}{m^2} e^{-\gamma(t+t')} \frac{1}{2\gamma} [e^{2\gamma t} - 1]$$

$$= \frac{C}{2m^2\gamma} [e^{-\gamma(t'-t)} - e^{-\gamma(t+t')}]$$

$$\approx \frac{C}{2m^2\gamma} e^{-\gamma(t'-t)}$$

t, t' large

$|t - t'|$ small

in small

$$\langle v(t) v(t') \rangle = \frac{C}{2m^2\gamma} e^{-\gamma|t'-t|}$$

Date

Einstein:

$$\frac{1}{2} m \langle v^2 \rangle = \frac{1}{2} k_B T$$

$$\langle v^2 \rangle = \frac{k_B T}{m} = \langle v(t) v(t) \rangle = \frac{C}{2m^2\gamma}$$

$$\Rightarrow C = 2m\gamma k_B T$$

$$\langle R(t) R(t') \rangle = 2m\gamma k_B T \delta(t-t') \quad \rightarrow \text{fluctuation dissipation theorem}$$

random force must be related to the friction to reach equilibrium

mean-square displacement

$$\langle x^2(t) \rangle$$

$$x(t) = \int_0^t v(\tau) d\tau \quad \text{assume } x(0) = 0$$

$$\langle x(t)^2 \rangle = \left\langle \int_0^t v(\tau) d\tau \int_0^t v(\tau') d\tau' \right\rangle$$

translational invariance
↓

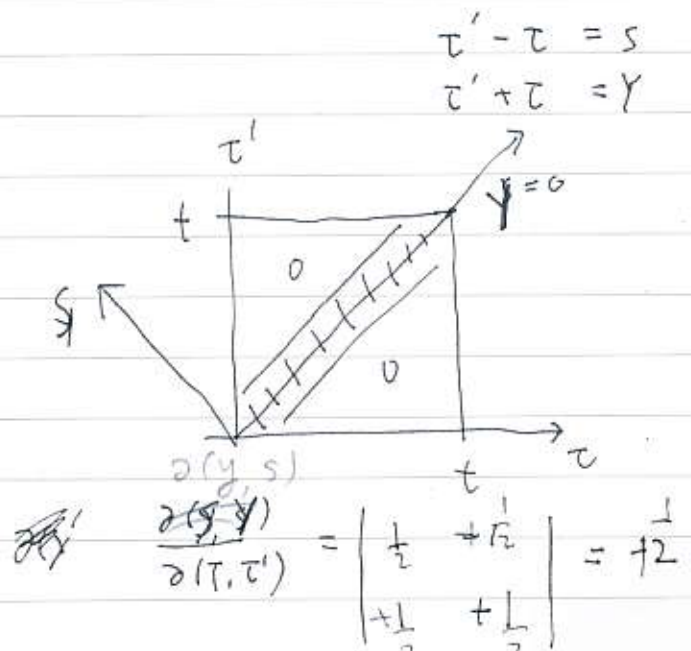
$$= \int_0^t d\tau \int_0^t d\tau' \langle v(\tau) v(\tau') \rangle = \int_0^t d\tau \int_0^t d\tau' \langle v(0) v(\tau'-\tau) \rangle$$

$$= \int_0^t d\tau \int_{-\tau}^{t-\tau} ds \langle v(0) v(s) \rangle$$

$$= \int dy ds \frac{\partial}{\partial y} \langle v(0) v(s) \rangle$$

$$= \frac{1}{2} \int_0^t dy \int_{-\infty}^{\infty} ds \langle v(0) v(s) \rangle$$

$$= t \int_{-\infty}^{\infty} ds \langle v(0) v(s) \rangle$$



$$\langle x(t)^2 \rangle \approx 2t \int_0^\infty \langle v(0) v(s) \rangle ds = 2D = 2Dt$$

$t \text{ large}$

$$\rightarrow D = \int_0^\infty \langle v(0) v(s) \rangle ds = \int_0^\infty \frac{k_B T}{m} e^{-\gamma s} ds$$

Diffusion constant

Green-Kubo formula

$\rightarrow \frac{k_B T}{m \gamma} = \frac{k_B T}{6 \pi \eta a}$

Einstein's relation

How to compute $\langle v(t_1) v(t_2) \dots v(t_n) \rangle$?

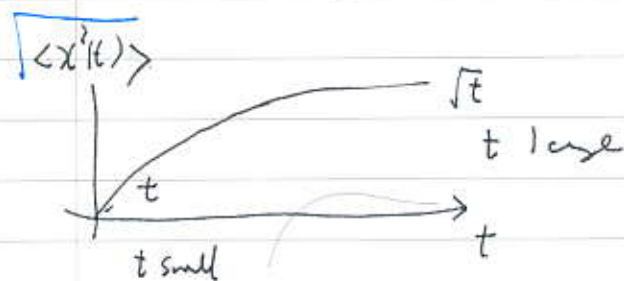
Need assumptions about high-order correlations of R
the simplest is gaussian process

$$\langle R(t_1) \dots R(t_n) \rangle = 0$$

if n is odd

$$\langle R(t_1) \dots R(t_n) \rangle = \sum_{\text{all possible pairings}} \langle \dots \rangle$$

(like Wick theorem)



Green-Kubo-type formula for transport coefficient

self-diffusion constant $D = \int_0^\infty \langle v(0) v(t) \rangle_{eq} dt$ $j = -D \nabla n$

Thermal conductivity $\kappa = \dots \int_0^\infty \langle j(0) j(t) \rangle_{eq} dt$ $\vec{j} = -\kappa \nabla T$
electric σ $\vec{j} = \sigma \vec{E}$

heat current

transport coefficients in nonequilibrium situation
(but near equilibrium (linear response regime))

is related to equilibrium correlation functions.

A more careful derivation

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(Uhlenbeck, Ornstein process)

$$\frac{dv}{dt} = -\gamma v + \frac{R(t)}{m}$$

$$\frac{dv}{dt} = -\gamma v \rightarrow v(t) = A e^{-\gamma t}$$

$$v'(t) = A(t) e^{-\gamma t} + A'(t) e^{-\gamma t} = -\gamma A e^{-\gamma t} + \frac{R(t)}{m}$$

$$A'(t) = \frac{R(t)}{m} e^{\gamma t}$$

$$A(t) = A(0) + \int_0^t \frac{R(\tau)}{m} e^{\gamma \tau} d\tau$$

$$v(t) = \left[v(0) + \int_0^t \frac{R(\tau)}{m} e^{\gamma \tau} d\tau \right] e^{-\gamma t}$$

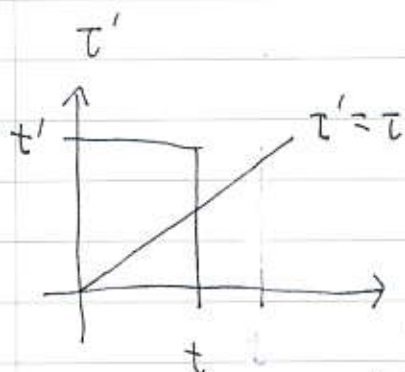
$$= v(0) e^{-\gamma t} + \int_0^t \frac{R(\tau)}{m} e^{-\gamma(t-\tau)} d\tau$$

assuming initial
velocity is dir δ
 $\langle v(0) | R(\tau) \rangle = 0$

$$\langle v(t) v(t') \rangle = v(0)^2 e^{-\gamma(t+t')}$$

$$+ \int_0^t d\tau \frac{e^{-\gamma(t-\tau)}}{m} \int_0^{t'} d\tau' \frac{e^{-\gamma(t'-\tau')}}{m} \langle R(\tau) R(\tau') \rangle$$

$$= v(0)^2 e^{-\gamma(t+t')} + \int_0^t d\tau \frac{e^{-\gamma(t-\tau)}}{m} \int_0^{t'} d\tau' \frac{e^{-\gamma(t'-\tau')}}{m} C \delta(\tau - \tau')$$



let assume $t' > t$

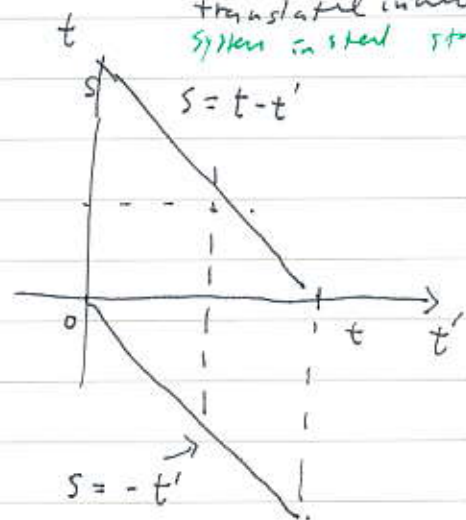
$$= v(0)^2 e^{-\gamma(t+t')} + \int_0^t d\tau \frac{e^{-\gamma(t-\tau)}}{m} \int_0^{t'} d\tau' \frac{e^{-\gamma(t'-\tau')}}{m} e^{-\gamma(t'-\tau)}$$

$$= v(0)^2 e^{-\gamma(t+t')} + \frac{1}{m^2} e^{-\gamma(t+t')} \int_0^t e^{-\gamma \tau} d\tau$$

$$\begin{aligned}
 &= v(0) e^{-\gamma(t+t')} + \frac{C}{m^2} e^{-\gamma(t+t')} \int_0^t e^{2\gamma\tau} d\tau \\
 &= v(0) e^{-\gamma(t+t')} + \frac{C}{m^2} e^{-\gamma(t+t')} \left[\frac{1}{2\gamma} (e^{2\gamma t} - 1) \right] \\
 &= \overset{2}{v(0)} e^{-\gamma(t+t')} + \frac{C}{2\gamma m^2} [e^{-\gamma(t'-t)} - e^{-\gamma(t+t')}] \quad t' > t
 \end{aligned}$$

in a stationary process t, t' large large $t - t'$ small

$$\begin{aligned}
 x(t) &= \int_0^t v(t') dt' \quad x(t=0) = 0 \quad t'' - t' = s \\
 \langle x^2(t) \rangle &= \left\langle \int_0^t v(t') dt' \int_0^t v(t'') dt'' \right\rangle = \int_0^t dt' \int_0^t dt'' \langle v(t') v(t'') \rangle \\
 &= \int_0^t dt' \int_0^t dt'' \langle v(0) v(s) \rangle \quad | \quad s = t'' - t' \\
 &= \int_0^t dt' \int_{-t'}^{t-t'} ds \langle v(0) v(s) \rangle \quad s = t'' - t' \\
 &= \int_0^t ds \int_0^{t-s} dt' \langle v(0) v(s) \rangle \\
 &\quad + \int_{-t}^0 ds \int_{-s}^0 dt' \langle v(0) v(s) \rangle \\
 &= \int_0^t ds (t-s) \langle v(0) v(s) \rangle + \int_{-t}^0 ds (t+s) \langle v(0) v(s) \rangle \quad s \rightarrow -s \\
 &= 2 \int_0^t ds (t-s) \langle v(0) v(s) \rangle \\
 t \rightarrow \infty \quad \langle v(0) v(s) \rangle &\text{ short-ranged. } \langle x^2(t) \rangle = 2t \int_0^t \langle v(0) v(s) \rangle ds \\
 &= 2tD
 \end{aligned}$$



using $\langle v(0) v(s) \rangle = \frac{c}{2m^2\gamma} e^{-\gamma|s|} = \frac{k_B T}{m} e^{-\gamma|s|}$

$$\langle x^2(t) \rangle = 2 \int_0^t ds (t-s) \frac{k_B T}{m} e^{-\gamma s}$$

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$$= \frac{2k_B T}{m} \left[t \int_0^t ds e^{-\gamma s} - \int_0^t s e^{-\gamma s} ds \right]$$

$$= \frac{2k_B T}{m} \left[\frac{-t}{\gamma} (e^{-\gamma t} - 1) - \frac{1}{\gamma} \int_0^t s ds e^{-\gamma s} \right]$$

$$= \frac{2k_B T}{m\gamma} \left[(1 - e^{-\gamma t}) t + \frac{s}{\gamma} e^{-\gamma s} \Big|_0^t - \int_0^t e^{-\gamma s} ds \right]$$

$$= \frac{2k_B T}{m\gamma} \left[t - \frac{e^{-\gamma t}}{\gamma} + \frac{t}{\gamma} e^{-\gamma t} + \frac{1}{\gamma} (e^{-\gamma t} - 1) \right]$$

$$= \frac{2k_B T}{m\gamma} \left(t - \frac{1}{\gamma} (1 - e^{-\gamma t}) \right)$$

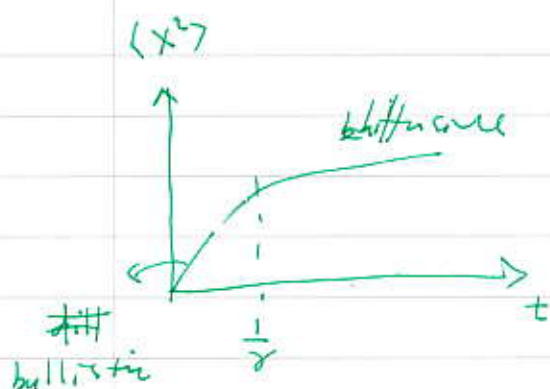
$$= \begin{cases} \frac{2k_B T}{m\gamma} t & t \rightarrow \infty \\ \frac{k_B T}{m} t^2 & t \rightarrow 0 \\ \uparrow \\ \langle v^2 \rangle t^2 \end{cases}$$

$$t \rightarrow \infty$$

$$t \rightarrow 0$$

$$D = \frac{k_B T}{m\gamma}$$

$$\gamma = \frac{1}{\lambda} \left(\gamma + \gamma + \frac{1}{2} (\gamma t)^2 \right) + \frac{1}{2} \gamma t^2$$



$$D = \frac{k_B T}{m\gamma} = \frac{k_B T}{6\pi a \eta}$$

$\uparrow$ viscosity
size of the molecule

Fourier Analysis

 $z(t)$ is realLet $z(t)$ periodic function in $0 \leq t \leq T$ ($T \rightarrow \infty$)

Fourier series

$$z(t) = \sum_{n=-\infty}^{\infty} a_n e^{i\omega_n t}$$

ω_n complex number

$$\omega_n = \frac{2\pi n}{T}$$

$$n = 0, \pm 1, \pm 2, \dots$$

$$a_n = \frac{1}{T} \int_0^T z(t) e^{-i\omega_n t} dt$$

go over to Fourier integral by $T \rightarrow \infty$, $\Delta\omega = \frac{2\pi}{T}$

$$z(t) = \int_{-\infty}^{\infty} a(\omega) e^{-i\omega t} \frac{d\omega}{2\pi}$$

change convention of Fourier transform

$$a_n \rightarrow a(\omega) \frac{2\pi}{T}$$

$$a(\omega) = \int_{-\infty}^{\infty} z(t) e^{+i\omega t} dt$$

since $z(t)$ is real

$$a^*(\omega) = a(-\omega)$$

Wiener-Khinchine theorem.

Power spectrum in ω -space is correlation in time space

correlation function

$$\phi(t) = \langle z(t_0) z(t_0 + t) \rangle$$

$$\text{assuming } \langle z(t) \rangle \equiv 0$$

② translational invariant

consider $I(\omega) =$

$$I(\omega) = \int_{-\infty}^{\infty} \phi(t) e^{+i\omega t} dt$$

$$z(t') = \int_{-\infty}^{\infty} a(\omega) e^{i\omega t'} d\omega$$

$$= \int_{-\infty}^{\infty} dt e^{+i\omega t} \langle z(t_0) z(t_0 + t) \rangle$$

$$= \int_{-\infty}^{\infty} dt e^{+i\omega t} \left\langle \int_{-\infty}^{\infty} a(\omega') e^{-i\omega' t_0} \frac{d\omega'}{2\pi} \int_{-\infty}^{\infty} a(\omega'') e^{-i\omega''(t_0 + t)} \frac{d\omega''}{2\pi} \right\rangle$$

$$\langle a(\omega) a^*(\omega') \rangle = \left\langle \left(\frac{1}{2\pi} \int_{-\infty}^{\infty} z(t) e^{+i\omega t} dt \right) \left(\frac{1}{2\pi} \int_{-\infty}^{\infty} z(t') e^{-i\omega' t'} dt' \right) \right\rangle$$

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$$= \left(\frac{1}{2\pi} \right)^2 \int dt \int dt' \langle z(t) z(t') \rangle e^{-i(\omega' t' - \omega t)}$$

Let $t' - t = t''$
 $t - t' = -t''$

$$= \left(\frac{1}{2\pi} \right)^2 \int dt \int dt'' \phi(t'') e^{-i(\omega' t'' + \omega' t - \omega t)}$$

$$= \int_{-\infty}^{\infty} \phi(t'') e^{+i\omega' t''} \left(\frac{1}{2\pi} \int_{-\infty}^{\infty} dt e^{-i(\omega' - \omega)t} \right)$$

$$= I(\omega') \delta(\omega' - \omega) \cdot 2\pi \quad \leftarrow \text{different } \omega \text{ are uncorrelated!}$$

$$= I(\omega) \delta(\omega - \omega') \cdot 2\pi$$

$$\rightarrow \langle |a(\omega)|^2 \rangle = I(\omega) \frac{T}{2\pi}$$

white noise $\langle R(t) R(t') \rangle = C \delta(t - t')$

$T \rightarrow \infty$

$$\text{Let } \tilde{R}(\omega) = \left(\frac{1}{2\pi} \right) \int_{-\infty}^{\infty} R(t) e^{+i\omega t} dt$$

using above result.

$$\langle \tilde{R}(\omega) \tilde{R}^*(\omega') \rangle = I_R(\omega) \delta(\omega - \omega') \cdot 2\pi$$

power spectrum of $\delta(t - t')$ correlates noise in ω -space is

$$I_R(\omega) = \left(\frac{1}{2\pi} \right) \int_{-\infty}^{\infty} C \delta(t) e^{+i\omega t} dt = \frac{C}{2\pi} \quad \text{in dpt of } \omega.$$

Is the Langevin Eq still valid if $\tilde{R}$ is not white noise? $\rightarrow$ NO

Colored noise

"Generalized Langevin Eq. gives colored noise"

fluctuation -
Dissipator
means

$$\frac{dV}{dt} = - \int_{-\infty}^t \gamma(t-t') V(t') dt' + \frac{R(t)}{m}$$

↑
memory effect

$$\langle R(t) R(t') \rangle = m K T \gamma(t-t')$$

$$V(t) = \int_{-\infty}^{\infty} \tilde{V}(\omega) e^{-i\omega t} \frac{d\omega}{2\pi}$$

$$R(t) = \int_{-\infty}^{\infty} \tilde{R}(\omega) e^{-i\omega t} \frac{d\omega}{2\pi}$$

similarly for $\tilde{V}(\omega)$

$$\int_{-\infty}^{\infty} \tilde{V}(\omega) \tilde{V}(\omega') e^{-i\omega t} \frac{d\omega}{2\pi} = - \int_{-\infty}^t \gamma(t-t') \int_{-\infty}^{\infty} \tilde{V}(\omega) e^{-i\omega t'} \frac{d\omega}{2\pi} dt' = \frac{1}{m} \int_{-\infty}^{\infty} \tilde{R}(\omega) e^{-i\omega t} \frac{d\omega}{2\pi}$$

$$= - \int_{-\infty}^t \gamma(t-t') e^{i\omega t'} dt' \tilde{V}(\omega) d\omega =$$

$t-t' = t'' \quad t' = t-t''$

$$= - \int_{+\infty}^0 d(-t'') e^{i\omega(t-t'')} \gamma(t'') \tilde{V}(\omega) d\omega$$

$$= \int_{-\infty}^{\infty} d\omega \tilde{V}(\omega) \left(- \int_0^{\infty} \gamma(t'') e^{-i\omega t''} dt'' \right) e^{i\omega t}$$

remember $\gamma(t) = \begin{cases} \gamma(t) & t \geq 0 \\ 0 & t < 0 \end{cases}$

define $\gamma(t) = 0$ if $t < 0$

$$= \int_{-\infty}^{\infty} d\omega [-\tilde{V}(\omega) \tilde{\gamma}(\omega)] e^{i\omega t}$$

$$\rightarrow \tilde{V}(\omega) (-i\omega) = -\tilde{V}(\omega) \tilde{\gamma}(\omega) + \frac{\tilde{R}(\omega)}{m}$$

$$\rightarrow \tilde{V}(\omega) = \frac{\tilde{R}(\omega)/m}{-i\omega + \tilde{\gamma}(\omega)}$$

$$I_v[\omega] = \int \langle v(t) v(t) \rangle e^{i\omega t} dt$$

$$I_v[\omega] = \frac{1}{m^2} \frac{I_R[\omega]}{(-i\omega + \gamma)^2}$$

by Wiener-Khinchine theorem

$$I_R[\omega] = m k_B T \tilde{\gamma}[\omega]$$

Fouri transform of $\gamma(t)$ not the same as that in damped kernel

problem of sedimentation / charged particle in a fluid

$$m \frac{dv}{dt} = -m\gamma v + f + R(t)$$

mobility

$$-m\gamma \langle v \rangle + f = 0$$

$$\langle v \rangle = \frac{1}{m\gamma} f = \mu f$$

$$\mu = \frac{1}{m\gamma}$$

$$D = \frac{k_B T}{m\gamma}$$

$$\text{so } \frac{D}{\mu} = k_B T \leftarrow \text{"Einstein relation"}$$

$$\text{Gauss distribution } \ln \phi(x) = \langle x \rangle + \frac{1}{2} \langle x^2 \rangle_c$$

How to solve the stochastic differential equation numerically?

$$m \frac{dv}{dt} = F + R(t)$$

moment generating function

$$\phi(\lambda) = \langle e^{\lambda x} \rangle = \sum \frac{\lambda^n}{n!} \langle x^n \rangle_c$$

$$\ln \phi(\lambda) = \sum \frac{\lambda^n}{n!} \langle x^n \rangle_c$$

$$\langle x \rangle_c = \langle x \rangle$$

$$\langle x^2 \rangle_c = \langle x^2 \rangle - \langle x \rangle^2$$

$$\rightarrow m dv = F dt + R(t) dt$$

$$m(v(t+h) - v(t)) = Fh + \int_0^h R(t) dt$$

$$= F \cdot h + \xi$$

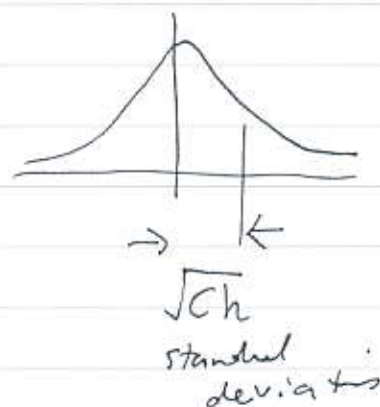
$$\xi = \int_0^h R(t) dt$$

$$\langle \xi \rangle = \int_0^h \langle R(t) \rangle dt = 0$$

$$\langle \xi^2 \rangle = \left\langle \int_0^h R(t) dt \int_0^h R(t') dt' \right\rangle = \int_0^h dt \int_0^h dt' \langle R(t) R(t') \rangle$$

$$\langle p^2 \rangle = \int_0^h dt \int_0^h dt' C \delta(t-t') = Ch$$

at each time-step h , generate a gaussian random number
 ξ with mean zero, variance Ch . $P(\xi)$



strong damping case (Smoluchowski?)

$$m \frac{dv}{dt} = -k v + R(t)$$

$$m = 0$$

k large

$$v = \frac{R(t)}{k} = \frac{dx}{dt}$$

$$x(t) = \frac{1}{k} \int_0^t R(t') dt'$$

$$\langle x(t) x(t') \rangle = \frac{1}{k^2} \left\langle \int_0^t R(t_1) dt_1 \int_0^{t'} R(t_2) dt_2 \right\rangle$$

$$= \frac{1}{k^2} \int dt_1 \int dt_2 C \delta(t_1 - t_2) = \frac{C}{k^2} \min(t, t')$$

$$\text{if } t = t' \quad \langle x(t)^2 \rangle = \frac{C}{k^2} t = \frac{2m\gamma k_B T}{(m\gamma)^2} t$$

Gaussian process

$$= 2 \left(\frac{k_B T}{m\gamma} \right) t$$

$$\int_{t_1}^{t_2} R(t) dt = W : W \text{ is a rand gaussian variable with zero mean and variance } t_2 - t_1$$

$$C = 2m\gamma k_B T$$

$$\cancel{2m\gamma k_B T}$$

$$= 2 \frac{k_B T}{\gamma} t$$